Temperature-Dependent Retention of Molten Lauric Acid in a Rice Bran Wax Matrix for Photovoltaic Thermal Management

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Draft. Items in red brackets await data (continuous-soak control, DSC, FTIR, SEM, field campaign) or a detail from the lab record.

Abstract

Photovoltaic (PV) modules lose about 0.43 % of their power per °C, and roof-integrated modules, with no rear ventilation, run hottest. A phase-change material (PCM) behind the module can cap the peak passively, but once molten it leaks. We measured how much molten lauric acid (melting point 43.8 °C) a network of rice bran wax, a rice-milling by-product, retains, and up to what temperature. Seven composites of 20–80 wt % lauric acid, cast under one thermal history, were held for two hours at 50, 60 and 70 °C (n = 3) against a pre-registered ≤ 2 wt % loss criterion. Every composition passed at 50 °C. At 60 °C the retention limit was 43 wt % (95 % CI 41–46); at 70 °C none passed and the 70–80 wt % specimens collapsed. Melting onset matched pure lauric acid, so no useful eutectic formed, and retention failed well below the ≈ 75 °C liquidus. After 25 melt–freeze cycles at 60 °C, the 40 wt % composite (C40) lost 16.1 %, eleven times its two-hour loss. Outdoors in Al Khobar over three days, 1 cm slabs behind insulated-back mini panels lowered the daily peak rear temperature by 6.5 °C (C40) and 6.7 °C (wax alone) on average, above the pre-registered 2 °C threshold, although with three days every 95 % confidence interval included zero. C40 lost 2.1 % of its mass outdoors. The wax holds molten acid well at 50 °C, but in the PV operating range retention depends strongly on temperature and time.

[PENDING — SOAK and DSC: add one sentence each when results arrive, keeping ≤ 250 words.]

Keywords: phase change material, shape stabilisation, rice bran wax, lauric acid, oleogel, leakage, building-integrated photovoltaics, waste valorisation

1. Introduction

A photovoltaic module turns most of the sunlight it absorbs into heat, and its output falls as it warms. Crystalline-silicon modules are rated at about **–0.43 % per °C** of maximum power above 25 °C; on a North Indian rooftop, cell temperatures peaked at **64.0 °C** and temperature alone cost **9.6 %** of output (Yadav et al., 2025). Northern Pakistan is similar: at Taxila, near the study site, hourly-average module temperatures ran at **34–50 °C** through May–July (Ali et al., 2016), and a reference module at Mirpur reached **64 °C** at 960 W/m² (Amber et al., 2021). A clear-sky rear temperature of **55–60 °C**, the range used throughout this work, is therefore routine, and it falls on a fast-growing fleet: Pakistan added 583 MW of net-metered rooftop PV in 2022–23 and 1,181 MW in 2023–24 (REN21, 2025).

Roof-integrated modules are the worst case, and the one this study targets. A module built into a roof or façade (building-integrated PV, BIPV) has its rear closed against the building. Sandia's module-temperature model puts an insulated-back module about **44 °C** above ambient at 800 W/m² and 2 m/s wind, against about **20 °C** on an open rack (King et al., 2004). Buildings are also where PV concentrates: about **75 %** of installed PV in Germany is on buildings (Fraunhofer ISE, n.d.). Rear ventilation and fins are unavailable when the rear is the building envelope, which is why PCM regulation of PV began with BIPV (Huang et al., 2004, 2006; Hasan et al., 2010).

A **phase-change material (PCM)** melting near the operating temperature absorbs latent heat at almost constant temperature and releases it overnight. Reported PV temperature reductions are mostly **5–20 °C**,

with efficiency gains of up to **16 %** (Sharaf et al., 2022). A recent review names the barriers to practical use as low conductivity, limited storage, **leakage** and price (El Kassar et al., 2025). This work addresses leakage. The standard remedy, shape stabilisation, uses either a rigid porous carrier or a gelling network (§2.2); the second is far less explored and is the one examined here.

Rice bran wax (RBW) is a by-product of rice-bran-oil refining. Pakistan, the world's ninth-largest rice producer (about 14.8 Mt of paddy in 2023; FAO, 2025), has it as a domestic residue. RBW gels vegetable oils at a few weight per cent and melts at **77–79 °C**, ΔH = 190.5 J/g (Dassanayake et al., 2009), far above the PV window, so its network could stay solid while a lower-melting PCM inside it does the buffering. That PCM is **lauric acid**, which melts at **43.8 °C** (Lide, 2009; DSC studies report 43.3–43.9 °C). Its ΔH is taken as **191 J/g** in the pre-registered predictions, while independent DSC puts it nearer **178–184 J/g** (Fan et al., 2022; Desgrosseilliers et al., 2013), so H3 is likely 4–7 % high. The material system was fixed first: the question is whether an RBW network can retain a molten *fatty acid*, and lauric acid is the fatty acid that melts inside the window.

This study prepares seven RBW / lauric-acid composites, measures how much molten acid each retains after a fixed hold at three temperatures, reports the retention limit against predictions recorded in advance, and tests the composite behind roof-integrated mini panels outdoors. It addresses UN Sustainable Development Goals 7, 12 and 13.

1.1 The research programme and its predictions

The project is a **first characterisation**, competing on experimental control rather than fabrication quality. Three predictions were recorded **before** any retention measurement: an ideal-mixing (Schröder–van Laar) **liquidus of ≈ 67 °C at 50 wt % lauric acid** (66–68 °C across an assumed wax molar mass of 600–900 g/mol, a larger uncertainty than the enthalpy value); a **network service limit of 55–68 °C**; and an **in-window latent heat of ≈ 191 J/g × (lauric-acid mass fraction)**.

Five exploratory water-bath runs came first; masses were not recorded, so they are characterisation only. One appeared to show the composite fully liquid at **48 °C**; a properly mixed replicate placed the liquidus near **75 °C**, and the 48 °C reading, traced to an incompletely mixed sample, was withdrawn with everything inferred from it (§6). That is why the preparation below is fixed tightly.

1.2 Objective and hypotheses

Primary objective. Measure the **retention limit X(T)**, the maximum lauric-acid mass fraction retained at ≤ **2 wt % mass loss** after a two-hour hold, at **50, 60 and 70 °C** across **20–80 wt % lauric acid**, and plot X(T) against temperature.

Secondary objectives. Locate the composite melting range and establish whether a *useful* eutectic forms; obtain latent heat versus composition by DSC; test for acid–wax chemical interaction by FTIR; relate retention to crystal-network density by SEM; and test the composite behind a module outdoors.

Hypotheses were stated with numeric predictions before measurement (H1–H7 with this paper; H7a–H7d in the field pre-registration of 14 October 2026, quoted as written):

#	Hypothesis	Prediction as stated	Falsified if
H1	Retention depends on loading	≤ 2 wt % loss at 50, 60, 70 °C up to a threshold lauric-acid loading	No composition meets ≤ 2 wt % at any temperature, or all do
H2	The network has a service limit	Retention ceases within **55–68 °C**	Retention at 60 and 70 °C is statistically indistinguishable from 50 °C across all compositions
H3	Latent heat scales with acid fraction	ΔH ≈ 191 J/g × w(LA), roughly linear	DSC enthalpy departs from linearity beyond scatter

#	Hypothesis	Prediction as stated	Falsified if
H4	Cycling stability and nucleation	Melting onset, enthalpy and retention drift **< 5 %** over 20–25 melt–freeze cycles, with no new FTIR bands; and the composite crystallises with **less supercooling** than pure lauric acid at matched DSC scale and cooling rate, wax crystals acting as heterogeneous nucleation sites	Drift ≥ 5 %, new FTIR bands appear, or composite crystallisation onset ≤ neat-acid onset
H5	No chemical interaction	FTIR ≈ superposition of components; no carbonyl shift	A reproducible carbonyl shift appears
H6	Structure controls property	Compositions failing the leak test show lower SEM crystal-network density and/or shorter mean needle length than those passing; density coarsens after 25 cycles (ImageJ, n ≥ 30 features)	No correlation across the composition series
H7	Field cooling	The optimised composite reduces back-surface temperature by **≥ 2 °C** against a matched bare panel under October conditions	ΔT < 2 °C, or indistinguishable from zero after crossover correction
H7a	The composite runs **cooler than the plain RBW slab** of the same thickness. The difference is the contribution of melting (latent heat) over and above added mass	—	Composite is not cooler than RBW (95 % CI includes 0)
H7b	C40 + graphite runs cooler than C40	—	95 % CI of (graphite − C40) includes 0
H7c	The composite melts by day and re-solidifies by night (outer-face sensors, between slab and board)	—	Outer face never passes 43 °C, or stays above it before dawn
H7d	Each slab loses **≤ 2 wt %** over the 4 days	—	Any slab loses > 2 wt %

Notes on the field rows. The field pre-registration falsifies H7 if "Mean ΔT < 2 °C". The stricter H7 condition above assumed a crossover design; the design as run had no crossover (§3.6.2), so that correction cannot be made and the zero test is applied without it. H7a was dropped in Version 2 and reinstated on 15 October when a wax-only slab was made. H7c was not tested, because no outer-face sensors were fitted. The campaign ran three days, not four.

Not hypothesised: electrical efficiency gain. The expected 0.8–2 % relative gain is smaller than the power scatter of unmatched mini panels, and no irradiance was measured on site, so efficiency is reported only as a derived quantity (temperature coefficient × ΔT).

The central hypothesis (H1–H2) is **refuted** if retention at 60 °C and 70 °C is statistically indistinguishable

from retention at 50 °C across every composition.

2. Literature review

Background is treated thematically (§2.1–2.2); the directly relevant prior work is then compared in one table (§2.3), and the gap stated (§2.4).

2.1 PV thermal loss and the case for passive PCM cooling

A relative power loss of about **0.45 % per °C** above 25 °C is the commonly quoted figure (Stropnik & Stritih, 2016, as cited in Durez et al., 2023). Yadav et al. (2025) give the realised effect for a 500 Wp North Indian array through one June: 64.0 °C peak cell temperature and 12.0 % average daily efficiency loss, 9.6 % of it from temperature alone and 20.4 % with soiling, which accumulated at 0.43 % per day. (The paper uses 0.43 both for the rated %/°C coefficient and for the daily soiling rate; the two must not be conflated.) Active cooling recovers more but adds pumps, power draw and maintenance. Among passive options a PCM acts specifically on the mid-day **peak**, which is why it is adopted here.

2.2 PCM leakage and the two shape-stabilisation mechanisms

Organic PCMs, paraffins and fatty acids, are the practical near-ambient choice, and loss of the melt is their recurring deployment failure (Zalba et al., 2003; Sharma et al., 2009). Containment works either by **porous confinement**, in which capillarity holds the liquid in a rigid open-pore carrier, or by **network gelation**, in which a few weight per cent of a crystalline gelator builds a space-filling mesh. Their temperature dependence should differ in kind: a gelled network holds liquid only while it stays *continuous*, so as its crystals dissolve into the melt on heating, retention should fail progressively, possibly well below the temperature at which the last crystal disappears. Whether the two mechanisms' retention curves differ in *shape*, not merely position, is open, and this project's data address it.

2.3 Prior work

Table 2.1. Directly relevant prior work and how each differs from the present study.

Source	PCM / liquid	Matrix or container	Setting	Key finding	Difference from this work
Dassanayake et al. (2009)	RBW, bulk and oil organogel	—	material	RBW melts at 77–79 °C, ΔH ≈ 190.5 J/g; strong oil gelator through long-chain wax esters	Liquid is an oil; no storage or retention data
Wijarnprecha et al. (2018)	Rice bran oil	RBW, 0.5–25 wt %	food oleogel	Gels oil from **0.5 wt %**; interlinked wax needles up to 50 µm long	Oil, not a free fatty acid; texture, not storage; no leakage test
Zhou et al. (2025)	Soybean oil	RBW + beeswax	food oleogel	Eutectic behaviour between the two waxes	Partner is a wax; no storage or retention data

Source	PCM / liquid	Matrix or container	Setting	Key finding	Difference from this work
Barrera et al. (2020)	RBW in rice bran oil, 1:7 v/v	aluminium container	PV under a xenon lamp, 45 min	Melts at **30.06 °C**, ΔH **34.95 J/g**; efficiency 18.776 % vs 18.544 % bare	Oil carrier melts below the PV window and dilutes latent heat; efficiency, not retention
Wang et al. (2022)	Lauric/stearic eutectic, 31.2 °C at 7:3	1–15 wt % expanded graphite	material	Conductivity ×2.4–2.6 at 10–15 wt %; no leakage test	Rigid porous confinement, not a dissolving network
Sun et al. (2024)	Stearic acid	peanut-shell carbon	material	≈ 0.53 W/m·K (×3.2), low leakage	Capillary carrier; the *comparison* mechanism here
Sarı (2005); Fan et al. (2022)	Lauric-acid binary eutectics	—	material	Pure lauric acid **43.93 °C, 178.11 J/g**; eutectics stable on cycling	Basis for lauric-acid properties and H4; no wax matrix
Durez et al. (2023)	Commercial PCMs at 21, 35, 44 °C	modelled layer	PV model, Bahawalpur, Apr–Aug	21 °C PCM gives best seasonal efficiency	Deliberate deviation, argued below
Huang et al. (2004, 2006)	RT25 paraffin	finned aluminium container	BIPV, experiment + 2-D model	Limits temperature rise by > 30 °C vs a flat plate	Contained; 25 °C melting for a temperate climate; fins for conductivity
Hasan et al. (2010)	Five PCMs, 25 ± 4 °C	containers	BIPV, indoor	−18 °C for 30 min; −10 °C held 5 h at 1000 W/m²	Temperate melting point; contained; no containment measure
Aelenei et al. (2014)	commercial PCM	BIPV façade unit	building, field	Max. electrical and thermal efficiency 10 % and 12 %	System-level; PCM integrity not studied
Soares et al. (2020)	RT 22 HC paraffin	25 mm aluminium units	open rack, Coimbra, 5 August days	PCM modules ran **14–21 °C hotter**, made 3.3–6.5 % less energy	Open rack, where the units blocked rear cooling; the reason the field test here uses an insulated back

Barrera et al. (2020) is the only prior study of RBW as a PV PCM. Its authors added wax to rice bran oil, which melts near 15 °C, "to increase the melting point to be within the accepted range", and reached only 30 °C. Replacing the oil with a fatty acid addresses that limitation and raises a containment question an oil system never faced: can a wax network hold a *fatty-acid* melt? (*Publiscience* is a high-school student journal; the study is treated as motivating prior art.)

Deviation from Durez et al. (2023). Their model favours a 21 °C PCM over the 35 and 44 °C materials. This project uses a 44 °C-class PCM on three grounds. (i) Their model starts each simulation with the PCM solid and does not treat re-solidification, and a 21 °C material cannot refreeze on Bahawalpur summer nights (June mean minimum **29 °C**; Rashid et al., 2023), whereas lauric acid has about 15 °C of margin. (ii) They optimise seasonal efficiency; this work targets the mid-day peak, and a 21 °C PCM is molten by

mid-morning. (iii) The melting point here follows from the material question. Practice agrees: RT42, a ~42 °C paraffin, is the PCM mostly used in summer conditions (Sharaf et al., 2022) and among the most effective pure PCMs for PV, with a 14.4 % efficiency gain (El Kassar et al., 2025). Two features of the Durez model are adopted: its one-dimensional, conduction-only enthalpy formulation and its PCM conductivity of **0.2 W/m·K** (§3.9.6).

2.4 Gap and aims

- **No RBW / free-fatty-acid composite has been characterised** for thermal storage. A July 2026 search of ScienceDirect, ACS, PubMed Central, RSC Advances and Wiley (*rice bran wax*, *lauric acid*, *fatty acid*, *phase change material*, *eutectic*, *shape stabilis(z)ation*) found RBW only as an oil oleogelator and, in Barrera et al., in an oil carrier.
- **No retention-versus-temperature data exist for a natural-wax network holding a molten fatty acid**, to the best of our knowledge (the search was oriented toward phase-behaviour rather than leakage terms).
- **No hold-time series exists for a gelled-network composite**, though it is routine for other classes. Khalaf Alshammari et al. (2024) weighed hexadecane / LDPE composites hourly for 20 h at 50 °C, using the same $(M_0 - M_n)/M_0$ quantity as here; losses were 0.10–3.64 % after 1 h and flattened after about 14 h. An et al. (2022) applied ten 1 h holds at 60 °C to paraffin / expanded-graphite composites and stayed below 4 wt %.
- **Post-cycling assessment is mostly calorimetric.** Wadee et al. (2025) ran 700 cycles with DSC and FTIR and tested leakage separately by 500 h at 50 °C; Du et al. (2021, as cited in Zhou et al., 2026) report 500-cycle stability on calorimetric grounds. For a natural-wax / fatty-acid network neither measurement exists.
- **No container-free, bio-derived PCM has been tested on an insulated-back module.** BIPV–PCM work holds commercial ~25 °C PCMs in containers (Table 2.1); added to an open-rack module, a PCM unit made it hotter (Soares et al., 2020).

This study therefore measures X(T) for RBW / lauric-acid composites at 50, 60 and 70 °C, re-measures retention after ≈ 10.6 h of accumulated heating so that the limit carries its hold time (§4.5), and tests the composite without a container behind insulated-back modules, measuring both module temperature and slab mass. Testing PCM on BIPV is not new; testing a container-free, waste-derived composite there, with its retention measured, is. No beneficial eutectic is claimed (§4.3).

3. Research methodology

The core method is a **gravimetric retention (leak) test** in the form widely used for shape-stabilised PCMs (heated hold, filter-paper capture, leakage as $(M_0 - M_n)/M_0 \times 100$; Khalaf Alshammari et al., 2024; An et al., 2022). The two-way mass balance, sealed control and pre-declared decision rules were added for this study.

Variables.

Type	Variable
Independent	Composition (20, 30, 40, 50, 60, 70, 80 wt % lauric acid); hold temperature (50, 60, 70 °C); cycle number (0, 25); field station (bare, RBW, C40, C40 + graphite)
Dependent	Specimen mass loss (%), the retention measure; melting onset and k_{eff} dips; ΔH, onset and supercooling by DSC; FTIR band positions; SEM network density and needle length; module back-surface temperature; slab mass loss

Type	Variable
Controlled	Specimen geometry and mass range; thermal history (melt, mix, pour, cool, 24 h anneal); 2 h hold; 30 min cooling before weighing; balance warm-up and re-zero schedule; module type, tilt and orientation; insulated back, open-circuit leads and sensor position, identical on every module; slab size

All preparation was carried out in the Chemistry Laboratory, Cadet College Hasanabdal. DSC, FTIR and SEM are performed as a facility service at [GIKI Topi, Swabi — confirm].

3.1 Materials

Material	Supplier	Grade	Batch	Role
Lauric acid	Bioshop PK	99 % (supplier-stated)	—	PCM
Rice bran wax	Aogubio Co., Ltd.	99 % (supplier-stated)	PR 1798	Matrix / gelator

Neither grade was verified independently (§6). All compositions are quoted on a **mass basis** from the balance readings actually recorded.

3.2 Composite preparation

Seven compositions were prepared, **20–80 wt % lauric acid** in 10 wt % steps, balance RBW (Figure 3.1). The protocol was: melt the RBW at **85 ± 2 °C** until clear; add the acid in two or three portions; hold **20 min at 85 °C** with periodic stirring, never above 100 °C; settle 5 min; re-weigh (a loss > 0.5 % flags overheating); pour into pre-warmed moulds to uniform depth, setting aside ~10 g of each as a melting-point specimen; cool on one surface without forcing; anneal **24 h** at room temperature.

As run. Melting and holding used a **Memmert UN30** natural-convection oven at 85 °C, not a water bath, and every composition soaked for more than 1 h at 85 °C before pouring. Casting masses were recorded (for example, C20: 11.98 g lauric acid and 48.05 g RBW), and nine specimens were cast per composition. All seven are taken to have had the same 24 h anneal [CONFIRM — recorded as outstanding in the Session 6 log].

One casting session. Cooling rate sets crystal structure, which sets retention, so all compositions were cast in one session under identical cooling. **Recorded deviation:** the melt solidified within ~1 min, so the compositions were poured **one at a time, 20 → 80 wt %**, the rest held at 85 °C. The extra residence (≤ ~15 min, after a common > 1 h soak) is argued to be negligible, but because the order was monotonic with composition it is aliased with the composition trend (§6).

3.3 Retention (leak) test

3.3.1 Design

A full factorial of **7 compositions × 3 temperatures × 3 replicates = 63 specimens**, each used once.

- **Temperatures.** 50 °C is above lauric acid's melt but below significant wax dissolution; 60 °C is a representative module rear temperature; 70 °C approaches the ~75 °C composite liquidus. The set brackets failure rather than avoiding it.



Figure 3.1. Laboratory work. (a) Materials: rice bran wax (batch PR 1798, foil bag), lauric acid, filter paper, specimen dishes and the mini panels. (b) Molten compositions in the Memmert UN30 oven at 85 °C, with the silicone specimen moulds pre-warming below. (c) Water-bath heating-curve set-up (§3.4). (d) Weighing specimens on the 0.01 g balance (§3.3.5). Photographs: [credit line — confirm who took each photograph].

- **Blocking by oven run.** Each run holds one replicate of all seven compositions (plus two blank papers and one sealed control), giving **nine runs**. Run-to-run variation therefore enters the error bar, and compositions are always compared within a shared load.
- **Randomisation.** Shelf positions within each run, and specimen-to-cell assignment, were randomised and recorded.

If specimens ran short, replicates were to be protected at the expense of temperatures (a complete n = 3 curve at two temperatures before any n = 1 work at a third).

3.3.2 Specimens and geometry

Leakage is a surface process, so the surface-to-volume ratio must be constant. Specimens were cast in one silicone mould, as **regular pentagonal prisms** (nominal cavity 19 mm across, 17 mm deep; the protocol's "cube"). Spot-checks on 12 specimens measured 1.85–1.95 cm across and 1.5–1.8 cm high. Masses were **4.0–5.4 g, mean 4.73 ± 0.34 g (n = 63)**, against a nominal 7 g. Specimens with a visible void, crack or skin defect were excluded before any run.

3.3.3 Oven and temperature control

Holds used the **Memmert UN30** oven, which displays chamber temperature to 0.1 °C under PID control, so dial offset and temporal cycling are controlled by the instrument. Spatial variation is handled by using the centre of one mid-height shelf and by position randomisation; the manufacturer's spatial uniformity is

[CITE — UN30 datasheet temperature uniformity]. The air flap was set once. The displayed temperature was logged before loading, on door closure, on recovery to within 1 °C of setpoint (which starts the two-hour hold), every 30 min, and at unloading.

3.3.4 Filter paper and blanks

Whatman No. 1 paper carries ~6 % water, which it loses in a warm oven. The protocol dried all papers at 105 °C and stored them over silica gel; **as run, papers were used straight from the box**, and the moisture correction rests on **two blank papers per run** (the protocol asked for three when drying was omitted). Two paper batches were used (mean dry mass 0.83 g at 50 °C; 1.02–1.03 g at 60–70 °C); only each paper's own before-and-after difference enters any calculation.

3.3.5 Procedure

On a balance of 0.01 g readability [CONFIRM — make/model], warmed and re-zeroed every ten weighings, the dish (m_{d0}), paper (m_{p0}) and specimen (m_0) were each weighed **three times and averaged**. Each was assembled dish → paper → specimen. All positions were loaded in under 30 s with one door opening, and the oven was not opened during the two-hour hold. Afterwards, specimen, paper and dish were re-weighed by the same method (m_1, m_{p1}, m_{d1}); the sealed control was weighed before and after every run.

As run, no desiccator was used: specimens cooled **30 min on the bench**, and their temperature was not measured before weighing (§6). The papers were **not photographed**, so the planned stain-area channel was not obtained. Full handling rules are in Appendix B.

3.3.6 Calculations and controls

$$\text{Leakage}\,(\%) = \frac{m_0 - m_1}{m_0} \times 100 \qquad \text{(primary)}$$

$$\text{Transfer}\,(\%) = \frac{(m_{p1} - m_{p0} - \Delta m_{\text{blank}}) + (m_{d1} - m_{d0})}{m_0} \times 100 \qquad \text{(cross-check)}$$

$$\text{Closure}\,(\text{g}) = (m_0 + m_{p0} + m_{d0}) - (m_1 + m_{p1} + m_{d1}) \qquad \text{(unaccounted mass)}$$

Leakage is the primary result and, because the specimen is non-hygroscopic, independent of paper moisture. **Transfer** recovers the lost mass on paper and dish (liquid runs down the prism sides past the paper edge, so the dish is weighed too). **Closure** is mass that left as vapour or was spilled. A **sealed control** (one specimen in a screw-cap vial, re-used across all runs) bounds evaporation directly. Whether two-hour leakage has plateaued is addressed by the soak arm of §3.5.

Pass criterion: mean Leakage ≤ **2 wt %** over three replicates, reported as mean ± SD.

3.3.7 Precision and decision rules

On 4.0–5.4 g specimens one balance increment is 0.19–0.24 % of m_0. The **limit of detection** is one increment (mean 0.21 %), the **limit of quantitation (LOQ)** three (mean **0.64 %**); values below LOQ are reported but not ranked. The 2 % criterion is 9–11 increments. Decision rules fixed before the first run (Appendix B) cover collapse (a specimen that cannot be lifted from its paper intact is a definitive **fail** with no leakage value), oven excursions (void the whole run), handling loss, defective specimens, blank drift and replicates (none dropped without a documented physical cause). The Leakage–Transfer threshold of 0.4 % is a diagnostic flag, never grounds for exclusion.

3.3.8 Analysis and headline output

X(T) is obtained by linear interpolation between the bracketing compositions:

$$X(T) = C_{\text{pass}} + (C_{\text{fail}} - C_{\text{pass}})\,\frac{2 - L_{\text{pass}}}{L_{\text{fail}} - L_{\text{pass}}}$$

where C_{pass} is the highest composition with mean leakage $L_{\text{pass}} \le 2$ % and C_{fail} the next one up. Its uncertainty is a bootstrap interval (2×10^4 resamples within cells).

Statistics. One-sided permutation tests (10^5 resamples, fixed seed) against ordered alternatives: a Jonckheere–Terpstra (JT) test for the composition trend at each temperature; a JT test stratified by composition for temperature, with stratified Mann–Whitney tests for each step and an exact sign test on cell means; and a within-cell permutation test for run-to-run variation. Collapsed specimens rank above every measured value but are never assigned a number. A two-way ANOVA is not claimed at this replication.

Pour-order control. The protocol specified recasting 20 and 80 wt % in reverse order and testing at 60 °C [Confirm whether the recast was run. If not, move it to §8 and keep the argued basis in §6.]

3.4 Supporting thermal characterisation

Melting behaviour is characterised by **indirect water-bath heating** with one sensor in the sample and one in the bath (Figure 3.1c). For purely sensible heating the sample's rate is proportional to the driving difference, so

$$k_{\text{eff}} = \frac{\mathrm{d}T_{\text{sample}}/\mathrm{d}t}{T_{\text{bath}} - T_{\text{sample}}}$$

is roughly constant, and a **dip in k_{eff}** marks heat going into latent form, locating a transition too broad to show as a plateau. Over each band $[a, b]$ it is evaluated in integrated form, $k_{\text{eff}} = (T_j - T_i) / \int (T_{\text{bath}} - T_{\text{sample}})\,\mathrm{d}t$, by the trapezoid rule (`Data Analysis/paper_figures.py`). Runs start from a cold bath and heat continuously to complete melting. The water-bath composite runs lack recorded masses and are preliminary (§1.1).

3.5 Thermal cycling stability

A material that passes a two-hour hold has not been shown to survive a season, so H4 is tested by repeated melt–freeze cycling of **C40**, the highest loading that passes at 60 °C (§4.4), cast in the same mould [Casting batch: the original Session 6 casting, or a new one? A new casting needs its own 2 h baseline specimens.]

3.5.1 Cycling protocol

Arm	n	Treatment
CYC	3	25 cycles: ~25 min in the oven at 60 °C, then cooled on the bench until the surface is dull white (acid frozen), minimum 25 min
SOAK	2	Held continuously at 60 °C for the same total hot time as CYC
Baseline	3	The C40 / 60 °C cell of the retention matrix (2 h)

CYC differs from the baseline in both cycle count and hot time. SOAK matches CYC on hot time, so **CYC − SOAK isolates cycling** and **SOAK − baseline isolates time at temperature**. Specimens sit open on paper in weighed dishes, as in §3.3, with one blank paper per arm; evaporation over ~10 h is not negligible but is common to CYC and SOAK. Every hot leg is timed. Cycling and soak are distinguished only if **|CYC − SOAK| > 1 percentage point**; a smaller difference is "no resolvable difference". Onset and enthalpy drift are measured by DSC and new bands by FTIR on the cycled specimen against an unheated reference (§3.7) [confirm whether the cycled specimen was submitted]. FTIR also guards against a degradation product that thickens the melt and makes cycled leakage look better than it is.

3.5.2 Tilted long-cell settling assay (designed, not performed)

Solid RBW (0.96–0.99 g/cm³) is denser than molten lauric acid (~0.85 g/cm³ at 70 °C). Once the network dissolves on a hot afternoon, the remaining crystals can settle toward the low edge of a tilted container, leaving unsupported acid at the top of the panel, its hottest region. A horizontal filter-paper test cannot detect this; El Kassar et al. (2025) likewise recommend inspecting composite PCMs for "degradation or separation within the composite material". The assay designed for it (a sealed cell ≥ 300 mm long at 20–35° tilt, cycled with CYC, photographed hot, then sliced into thirds and the wax fraction measured) **was not run**, so no extrapolation from §3.3 to a tilted module is made (§6, §8).

3.6 Field evaluation

H7 was tested on a roof in **Al Khobar, Saudi Arabia** (26.2° N) on **15, 16 and 17 October 2026**, behind **insulated-back (roof-integrated) modules** (Figure 3.2). The planned site was Cadet College Hasanabdal. Al Khobar's October afternoons were hotter than the model's climate: modelled air 33–41 °C and ≈ 800 W/m² at noon (Open-Meteo), against 15–29 °C (§3.9.6).

3.6.1 Stations and slabs

Four frameless mini mono-crystalline panels of one model, each **1 W, 3 × 5 in (7.6 × 12.7 cm, 97 cm²)**, not IV-matched, formed four stations:

- **(A) bare**;
- **(B) rice bran wax alone**, the thermal-mass control: RBW melts only near 70–73 °C (§4.1), so it adds mass and sensible heat without latent heat, and C − B isolates the latent effect;
- **(C) C40**;
- **(D) C40 + graphite**, 4 g natural graphite per 100 g composite (≈ 3.8 wt %), from a second casting.

No container is used. Each slab is **14 × 7 × 1 cm** (measured after the campaign; the pre-registration assumed ≈ 2 cm). Masses before: RBW 98 g, C40 96 g, C40 + graphite 100 g. A slab covers ≈ **92 %** of the module rear (12.7 × 7 cm), overhangs one end by ≈ 1.3 cm and is notched around the two lead joints (Figure 3.2b). It is held between module and board by tape on the module edges; nothing is stuck to the slab.

Mounting. Every module, the bare control included, is taped to its own 20 mm expanded-polystyrene board, whose lower edge rests on a box and upper edge on a south-facing wall (tilt ≈ 35°). Every sensor is therefore covered, by a slab or by the bare station's board, so none is convectively cooled while another is insulated. Leads are open-circuit, ends taped separately; catch paper lies under every module.

Sensors. One DS18B20 is taped to each module back at the same measured position on all four and **sandwiched between the module and its slab** (on the bare station, between the module and its board), on all three days. Each therefore reads the module's rear surface where the slab presses against it. *Correction, 17 October 2026: earlier records, including the Day 1 note in the field pre-registration, place the sensor just below the slab's lower edge; the authors confirm it sat between module and slab throughout.* Sensors are identified

by ROM address; the mapping was confirmed by warming each probe on 16 October (S0 C40 + graphite, S1 C40, S2 RBW, S3 bare), correcting Day 1's file header.



Figure 3.2. Field rig. (a) The four stations on the roof, each module taped to its own polystyrene board, slab edges visible behind three of them. (b) The rear of a mini panel, showing the two lead joints the slabs are notched around. (c) The logger: an Arduino Uno and breadboard reading the four DS18B20 sensors over USB. Photographs taken by the authors.

3.6.2 Design and controls, as run

- **Rotation.** Each morning whole stations (board, slab and module) moved one position along the row, spreading position effects across days, but **panel and sensor bias stay tied to treatment**. The bare calibration run (all slabs off, stations moved every 15 min) was not available for analysis, so no panel correction is applied (§6).
- **Timing.** Day 1 logging began at ≈ 13:30, after solar noon (≈ 11:25); Day 2 at ≈ 11:47; Day 3 at 11:45, running to ≈ 18:50, past sunset (≈ 17:15). No pre-dawn logging. The stations stayed outdoors throughout the campaign, about 72 h in total. On Day 3 the row was checked every 30 min. The logger had no clock, so start times are recorded by hand.
- **Logging.** An Arduino Uno read the four sensors every 2 s; raw Serial Monitor transcripts are kept unchanged beside the clean files.
- **Weather.** No on-site pyranometer or ambient sensor was available. Irradiance, air temperature, wind and cloud come from the Open-Meteo API (Zippenfenig, 2023) at 15 min resolution, interpolated to each reading. These are **model values**, used to tell cloud from local shadow and to compare stations at matched sunlight.
- **Mass.** Slabs were weighed before the first day and after the last on a 1 g kitchen scale. The C40 + graphite slab partly collapsed on Day 2, was left in place for Day 3, and was weighed broken, without fragments stuck to its board.
- The field ΔT is compared against the **pre-registered model prediction** (§3.9.6, v2).
- **Not run.** A side-by-side oven hold of C40 and C40 + graphite specimens at 60 and 70 °C, under the §3.3 protocol, is needed to tell whether graphite itself weakens the network or the collapse was a field effect [PENDING — C40 vs C40 + graphite oven test].

3.7 Sample preparation for external instrumental analysis

DSC, FTIR (ATR) and SEM are performed by facility staff; the authors prepare, seal, label and submit the samples and analyse the returned data. The requested sample sets and programmes are in Appendix C. [Record from the facility report: instrument models, samples actually run and programme actually used.] In brief: **DSC** on raw RBW, pure lauric acid and C30/C50/C70 (all seven if time allows), loaded molten, heat–cool–heat 20 ⟶ 90 ⟶ 20 ⟶ 90 °C at 1–2 °C/min under N_2, with supercooling taken from the cooling segment at matched mass and rate; **FTIR** on the same series plus cycled C40, examining the carbonyl (≈ 1700 and ≈ 1735 cm^{-1}) and O–H regions; **SEM** on raw RBW, one low- and one high-acid composite, and uncycled and cycled C40, with network density and needle length measured in ImageJ (n ≥ 30 features).

Scope limit. A DSC pan holds 5–20 mg, and a milligram sample supercools more than a bulk one, so the

H4 supercooling comparison is valid at DSC scale only. No bulk T-history apparatus was built; the field evening cooling rates (§4.9.2) give partial evidence at real scale.

3.8 Sensor calibration

The four **DS18B20** sensors (12-bit, 0.0625 °C resolution, ±0.5 °C factory accuracy) were given a **one-point offset** in a stirred crushed-ice slurry, tips bundled, each reference the mean of 10 settled readings. Ice-bath means were **1.61, 1.39, 1.86 and 1.37 °C** (S0–S3; raw spread 0.49 °C). The bath sat above 0 °C, so the offsets are used **only to compare sensors with one another**. In an independent room-temperature water check, the raw spread of 0.41 °C fell to **0.11 °C** after correction; that residual is the sensor-to-sensor uncertainty used here. It was measured at ≈ 28 °C only: span (slope) differences at 40–70 °C are not corrected, and the planned boiling-water point was not run. The melting-point trials used a 1 °C glass thermometer and the retention holds the oven's display; the field data are the only results that use these sensors.

3.9 Data analysis and statistics

3.9.1 Retention series

As set out in §3.3.8. Scripts in `Data Analysis/` regenerate every figure and number from the workbook (`Data Analysis/leak_analysis_matrix.py` → `outputs/leak_matrix_results.txt`).

3.9.2 Heating curves

A single line fitted across a phase transition is meaningless, so each heating trial is split at its first melting observation into a **sensible ramp** and a **melt region**, and the melt-region slope is divided by that trial's own ramp (the cut-point is declared as a researcher degree of freedom). Where the bath was logged, the boundary-free k_{eff} analysis of §3.4 is used instead. These are time series with serially correlated residuals, so ordinary least-squares p-values are optimistic and are not presented as if observations were independent.

3.9.3 Uncertainty budget

Type A (replicate and day-to-day scatter) and Type B (±0.5 °C sensor accuracy; 0.01 g readability, $a/\sqrt{12}$ for a rectangular limit) are combined in quadrature. For a field difference, $u(\Delta T) = \sqrt{(u_1^2 + u_2^2)}$: 0.71 °C with uncalibrated sensors, falling to the 0.11 °C inter-sensor spread after the offset correction. The sensors therefore do not limit the field comparison; **panel and position bias** does, because with one module per treatment the design does not remove it (§3.6.2, §6). Propagating V, I, irradiance and area into efficiency gives an uncertainty larger than the expected effect, which is why efficiency is not hypothesised. Reported precision matches uncertainty throughout.

3.9.4 Field data

Ice offsets are applied per sensor, and each slab station is differenced against the bare station at the same moment.

Pre-registered main number. For each day, each slab station's peak is subtracted from the bare station's peak (Day 2's last 5 min, packing up, excluded). The three daily values are reported as mean ± SD with a t-based 95 % CI (n = 3). No other significance test is run; effects are reported as sizes with their conditions.

Post-hoc analyses, labelled as such. (i) A **findings window** of the first 90 min of each day, chosen after Day 2's data showed a shadow crossing the row from ≈ 95 min (modules dropped one after another at 0 % modelled cloud); on Day 3 the row lost direct sun from ≈ 180 min. **No reading is discarded**: full

records are plotted and reported. (ii) **Matched sunlight**: readings with modelled irradiance ≥ 600 W/m², shadow excluded, pooled across days. (iii) **Evening cooling rate** (10 min running mean) compared at equal module temperature after the row loses sun: a slab releasing latent heat on freezing should cool more slowly near 43–44 °C. Modelled cloud cover was ≤ 7 % on all days, so the pre-registered cloud criterion excluded no day. Analysis: `Data Analysis/field_analysis.py`; raw files in `Field Test Data/`.

3.9.5 Comparison against the pre-registered predictions

Measured DSC enthalpies are compared against the **191 J/g × *w*(LA)** line (zero intercept) and the measured liquidus against the ideal Schröder–van Laar curve. Departures are reported as findings with their direction; predictions are not adjusted after the fact.

3.9.6 Panel-scale thermal model and its pre-registered prediction

H7 concerns a panel, not a material. The two are bridged by a lumped two-node transient model (`Data Analysis/pcm_panel_model.py`), run **before** the field campaign so that its output is a falsifiable prediction. The module is one node; the composite is a second node handled by the **enthalpy method over its measured melting band** (solidus 43 °C, liquidus 75 °C). Heat leaves the front by wind convection and long-wave radiation to sky and ground; the rear either loses heat to open air (open rack) or conducts through a 20 mm polystyrene board (insulated back), and the same boundary is applied to the bare module and to the composite's outer face. Integration is explicit Euler at 2 s over three days of a synthetic October climate for northern Pakistan (Topi, near the planned Hasanabdal site: 15–29 °C, 800 W/m² clear-sky peak, 2 m/s wind). Equations are in Appendix A.

Table 3.1. Model parameters.

Symbol	Quantity	Value	Status
T_s, T_l	solidus, liquidus	43 °C, 75 °C	measured (§4.2, n = 1)
ΔH	latent heat of the composite	76 J/g	prediction for C40 (191 J/g × 0.40); v1 used 95 J/g
ρ	density	900 kg/m³	placeholder
c_p	specific heat	2.0 J/g·K	literature (Durez et al., 2023)
k	conductivity	0.2 W/m·K	literature (Durez et al., 2023)
L, A	layer thickness, module area	0.02 m, 0.10 m²	design
C''	module areal heat capacity	9000 J/m²·K	order-of-magnitude estimate
(τα)	transmittance–absorptance	0.90	assumed
h_b	back-face convection	8 W/m²·K	assumed, identical in both cases
ε_f, ε_b, ε_c	emissivity: glass, backsheet, composite	0.85, 0.85, 0.90	assumed
β	tilt	35°	design (field rig)
T_{sky}	clear-sky temperature	$0.0552\ T_a^{1.5}$	Swinbank (1963)
L_i, k_i	insulation board	20 mm, 0.035 W/m·K	design; typical expanded polystyrene
$V \rightarrow h_w$	wind → front convection	2 m/s → 13.2 W/m²·K	Durez et al. (2023), Eq. 1
η_{ref}, β	efficiency at 25 °C, coefficient	0.18, 0.0043 °C⁻¹	rated (Yadav et al., 2025)

Symbol	Quantity	Value	Status
Δt	timestep	2 s, 3 days	numerical (convergence in §4.10)

The model departs from Durez et al. (2023) deliberately: it runs **multi-day** and reports whether the PCM re-solidifies each night, models a **melting band** rather than a sharp melting point, and (from v2) includes **long-wave radiation** and a choice of **mounting**. ΔH, ρ, c_p and **k** are placeholders pending DSC and hot-wire measurement, with k the dominant unknown, so the model verifies **behaviour**, not magnitudes.

The prediction, stated in advance. Two versions exist, both recorded before any field data. Both are reported, and neither is revised after the campaign.

Version 1 (git commit 63b03fa; open rack, convection only, 0.1 m² module, 2 cm layer, ΔH = 95 J/g). At the placeholder conductivity the model predicts that the composite layer **fails H7, and fails it in the wrong direction**:

	Bare	With composite
Peak back-surface temperature	61.5 °C	**66.7 °C**
Peak ΔT	—	**−5.3 °C** (hotter)
Daylight mean ΔT	—	−3.8 °C
Mean efficiency change	—	−0.16 pts (−0.9 % relative)
H7 (≥ 2 °C drop)	—	**not met**

Version 2 (15 October 2026, before the campaign). The *design* changed to insulated-back modules, C40 and 2 cm slabs; the *model* gained long-wave radiation, which lowers the v1 open-rack bare peak from 61.5 °C to 47.8 °C, close to the ≈ 49 °C that Sandia's coefficients give for the same conditions (King et al., 2004). For the field configuration v2 predicts:

	Bare	C40, 2 cm	C40 + graphite, 2 cm (k 0.3–0.4)
Peak back-surface temperature	59.2 °C	57.3 °C	57.0–57.1 °C
Peak ΔT	—	**+1.9 °C** (cooler)	+2.1 to +2.2 °C
Daylight mean ΔT	—	+0.8 °C	+0.8 to +0.9 °C
Peak liquid fraction	—	0.25, solid again by every dawn	0.31–0.35
Peak composite temperature	—	51.0 °C	53–54 °C
H7 (≥ 2 °C drop)	—	**borderline: not met by 0.1 °C**	met by 0.1–0.2 °C

Three features of v2 matter. **Mounting decides the sign:** the same slab on an open rack runs 6.0 °C hotter than bare and never melts, while against insulation ($U \approx 1.5$ W/m²·K, against $k/L = 10$ W/m²·K for the slab) its storage dominates. **Conductivity barely matters:** on the insulated rear, k from 0.2 to 6 W/m·K moves peak ΔT only between +1.6 and +2.2 °C, so graphite should make no detectable difference. **About half the cooling is mass:** a wax-only slab is predicted at **+0.8 °C**, so C40 should run about 1.1 °C cooler than wax alone at peak.

The slabs were later measured at **1 cm**. The 2 cm prediction is the one tested; for reference only, the same model at 1 cm gives C40 **+0.8 °C** at peak (liquid fraction 0.40, composite peak 56 °C). The field ran in a hotter climate than the model (§3.6), so the comparison is made on **differences between stations**, and the prediction is not re-run on Al Khobar weather. Three outcomes were distinguished in advance:

the field ΔT matches, exceeds or falls short of the prediction. **The prediction is not revised after the campaign.**

3.10 Safety

Molten lauric acid at 70 °C burns skin: oven gloves for trays, forceps for specimens. The acid is a mild eye and skin irritant and warm acid gives off a strong vapour, so gloves, eye protection and ventilation were used, and all heated work was supervised by the chemistry teacher. Paper and solvent were kept clear of the oven elements. **Graphite powder** was weighed in ventilation with a dust mask, away from the balance and melt, and kept off the modules' bare lead ends, since a graphite bridge would short the module; the collapsed graphite slab was handled with gloves. Slabs were cut with a warm, dried knife, away from the hand. In the field, the frameless panels were carried by their edges; module voltage is a few volts and leads were open-circuit. The laptop, the only mains item, was kept shaded, raised off the roof and off wet ground, and taken inside above about 35 °C shade temperature. The team worked in shade where possible and carried water. Leaked composite was disposed of as solid waste with its catch paper, never down a drain.

4. Results

4.1 Pure component melting points

Both components were heated in the water bath of §3.4 and read from a 1 °C glass thermometer, which plateaued at **98 °C** in boiling water at the 450 m site (expected ≈ 98.5 °C), so no correction was applied.

Lauric acid

Table 4.1. Lauric acid, two heating trials (n = 2).

	Mass	Pre-melt ramp	Onset	Plateau	Clear-melt
Trial 1	4.95 g	1.33 °C/min	43 °C	**43.0 °C, held 4.0 min** (8 readings)	46.5 °C
Trial 2	5.12 g	2.13 °C/min	44 °C	none	52 °C

Melting onset is **43–44 °C**, matching the literature 43.8 °C within the thermometer's resolution, so the supplied acid is taken as validated.

The trials also give a methodological result. Trial 2's ramp was steeper by **+0.80 °C/min** (95 % CI 0.47–1.13; t(15) = 5.22, p = 0.0001; optimistic, §3.9.2). Through the transition, Trial 1's rate of rise fell to **35 %** of its sensible rate and Trial 2's only to **86 %**: Trial 2 outran the latent absorption. The onset moved by 1 °C between trials, the "fully melted" reading by **5.5 °C**. Mass does not explain this, since the heavier Trial 2 should have ramped more slowly. **Onsets and plateaus are robust; the visual clear-melt endpoint is not**, which is why transitions are read from arrests and k_{eff} dips throughout.

Rice bran wax

Table 4.2. Rice bran wax (batch PR 1798).

	Mass	Method	Transition	Note
Trial 1	4.79 g	plateau	**73 °C**, flat 5.25 min (18 readings)	clear-melt 84 °C

	Mass	Method	Transition	Note
Trial 2	4.45 g	plateau	apparent flat at 82 °C, 1.75 min	bath not logged; **not a melting arrest** (see Trial 3)
Trial 3	not recorded	bath-logged k_{eff}	**70–73 °C**, lowest k_{eff} of six bands	no visible plateau; k_{eff} highest at 79–83 °C

Table 4.3. Trial 3, k_{eff} by sample-temperature band (integrated form, §3.4).

Band (°C)	63–67	67–70	**70–73**	73–76	76–79	79–83
k_{eff} (min^{-1})	0.090	0.138	**0.058**	0.101	0.113	0.175

Two independent observables place the wax transition at **70–73 °C**: Trial 1's 5.25 min plateau, and Trial 3's lowest k_{eff} band (36 % below the next-lowest, 49 % below the median of the others), visible only once the rate is divided by the still-rising bath–sample difference. Trial 2's 82 °C flat did not reproduce: Trial 3 has its *highest* k_{eff} at 79–83 °C, so no latent heat was absorbed there, and the flat is attributed to heat input.

The transition lies **about 5 °C below** the 77–79 °C of Dassanayake et al. (2009). A thermometer offset is excluded by the boiling-point and lauric-acid checks. RBW is a mixture of long-chain esters whose distribution depends on source and refining, so a lower-melting fraction is plausible even at a stated 99 % grade. DSC on the raw wax is the arbiter (§4.6) [PENDING].

4.2 Composite melting behaviour (preliminary)

> **Preliminary.** Masses were not recorded for the water-bath composite runs, so their compositions are approximate. The quantified version is the composition-series melting test on specimens set aside at casting [PENDING — run / not run?] and the DSC series (§4.6).

The properly mixed run (≈ 55 wt % lauric acid, bath logged, heated continuously) melts over **43 ⟶ 75 °C**.

Table 4.4. Composite (≈ 55 wt % LA), k_{eff} by band, and relative to the run's maximum.

Band (°C)	38–43	**43–45**	**45–47**	**47–50**	50–57	57–67	67–75
k_{eff} (min^{-1})	0.091	**0.028**	**0.048**	**0.030**	0.171	0.497	0.201
relative	0.18	**0.06**	**0.10**	**0.06**	0.34	1.00	0.40

Melting proceeds in **two stages** (Table 4.4, Figure 4.3). From **43 to 50 °C the acid melts**: k_{eff} falls to 6–10 % of its maximum, so most incoming heat goes into latent form. From **50 to 75 °C the residual wax dissolves**, absorbing little heat but visibly: 90 % melted at 72 °C, 5 % solid at 73 °C, fully melted at **75 °C**. The bath was still 13 °C hotter at the end, so 75 °C is a genuine completion, not an equilibrium ceiling.

A second, lower-acid run was not a clean replicate (different mixing, pre-warmed bath, heat cut mid-run, stopped at 77 °C "almost" melted). Its main absorption moved to **54–58 °C** and its liquidus is bounded at ≥ **77 °C**, as a lower acid fraction should give. Both runs put the last solid in the **mid-70s °C**. The pre-registered ideal-mixing liquidus is **≈ 67 °C at 50 wt %**; the observed **≈ 75 °C at ≈ 55 wt %** is at least 8 °C higher, so the wax is *less* soluble in molten lauric acid than ideal mixing predicts (§5.2). Bulk latent heat is thus delivered at 43–50 °C, inside the PV window, while solid wax persists above it; whether that solid **retains** the liquid is a mechanical question answered in §4.4.

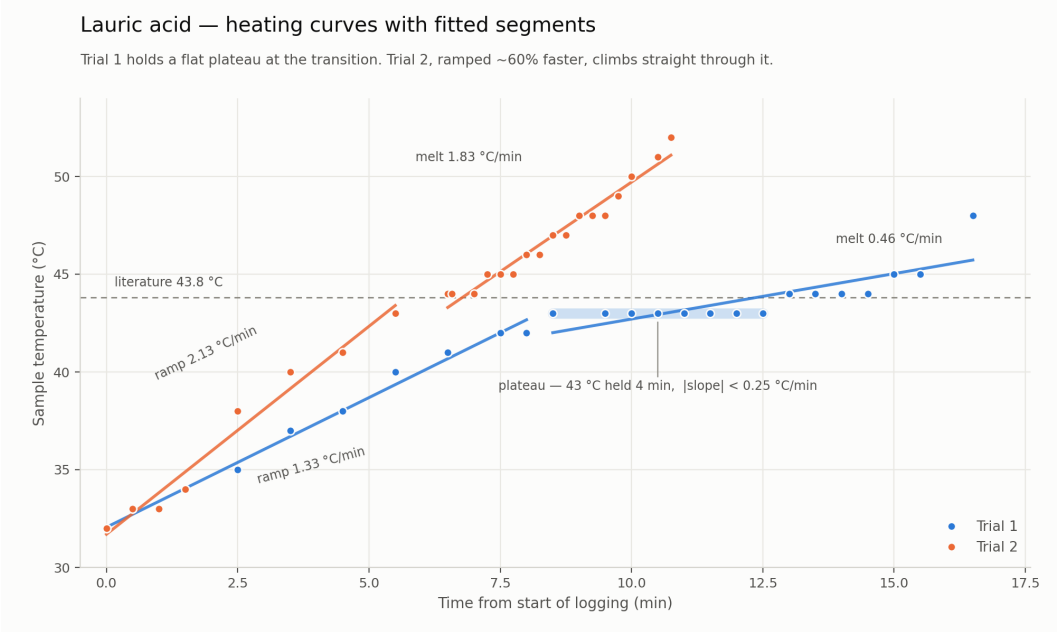


Figure 4.1. Lauric acid, Trials 1 and 2, with fitted pre-melt ramps and melt segments. Trial 1 holds 43.0 °C for 4.0 min; Trial 2, ramped about 60 % faster, climbs straight through. Dashed line: the 43.8 °C literature value.

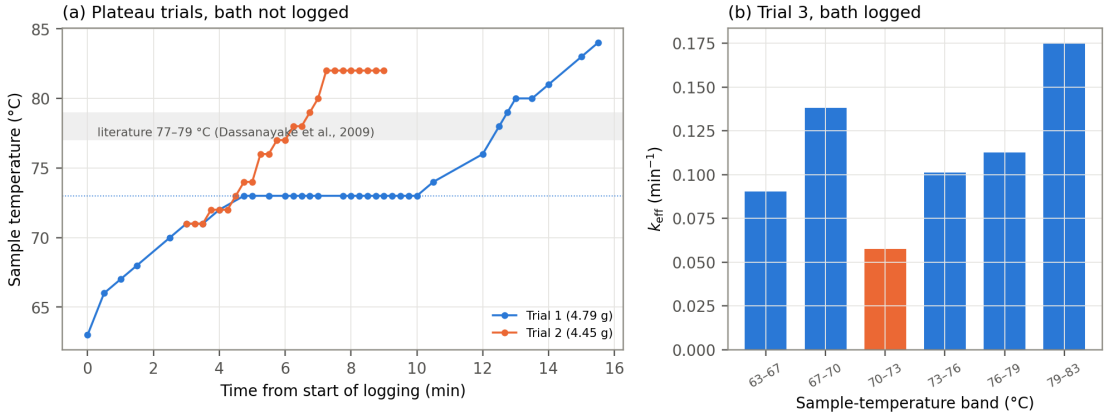


Figure 4.2. Rice bran wax. (a) Plateau Trials 1 and 2, bath not logged; grey band, the literature melting range. (b) Trial 3, bath logged: *k*eff by band; the 70–73 °C band (orange) is the lowest.

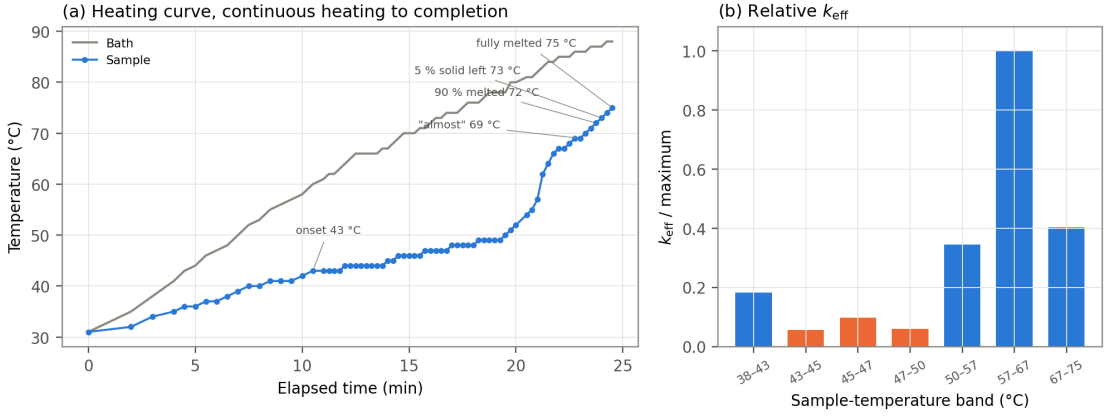


Figure 4.3. Properly mixed composite (≈ 55 wt % lauric acid). (a) Bath and sample temperatures with the visual milestones. (b) *k*eff by band relative to its maximum; the 43–50 °C bands (orange) are where the acid melts.

4.3 Absence of a useful eutectic

The ideal-mixing model put the eutectic at **43.72 °C**, indistinguishable from pure lauric acid. Composite onsets were **43, 43 and 44 °C** in three runs (one being the withdrawn, poorly mixed run, whose onset still reflects the acid). No melting-point depression is seen. A *useful* eutectic, in the fatty-acid sense of a melting point well below either component (lauric/stearic acid at 31.2 °C; Wang et al., 2022), does not form. The wax dilutes latent heat roughly in proportion to the acid fraction (≈ 95 J/g at 50 wt % on the 191 J/g × w rule, to be checked by DSC, §4.6). The wax's role is containment only.

4.4 Retention versus temperature

4.4.1 Completeness of the matrix

The full design was run: **63 specimens**, none voided. Initial masses were **4.0–5.4 g** (mean 4.73 ± 0.34 g), not the nominal 7 g. Six specimens **collapsed**, all replicates of 70 and 80 wt % at 70 °C, and are recorded as definitive fails without a leakage value (§3.3.7). The call was not borderline: the largest loss among the 57 intact specimens was **10.6 %**, the smallest among the collapsed **42.3 %**.

Three supporting items are incomplete. **Dish, sealed-control and 60/70 °C blank masses** are in the notebook but not yet in the analysed workbook [TRANSCRIBE], so the cross-check (§4.4.5) currently covers paper gain only. **Stain-area photographs** were not taken. **Long-hold behaviour** awaits the SOAK arm (§4.5) [PENDING]; until then all values here are **two-hour** leakage, not equilibrium values. The primary measurement, specimen mass loss, needs none of these.

4.4.2 Leakage across the matrix

Table 4.5. Leakage after a two-hour hold, % of initial specimen mass, mean ± SD (n = 3), range in brackets. **Bold** marks a mean that fails the ≤ 2 wt % criterion.

wt % LA	50 °C	60 °C	70 °C
20	0.1 ± 0.1 [0.0–0.2]	0.6 ± 0.6 [0.2–1.3]	**2.5 ± 0.7** [1.9–3.1]
30	0.3 ± 0.1 [0.2–0.4]	0.8 ± 0.3 [0.5–1.1]	**3.2 ± 0.5** [2.7–3.7]
40	0.4 ± 0.5 [0.0–1.0]	1.5 ± 0.3 [1.2–1.8]	**3.5 ± 0.6** [3.1–4.2]
50	0.3 ± 0.3 [0.0–0.6]	**3.2 ± 1.2** [2.3–4.6]	**7.2 ± 1.7** [5.6–8.9]
60	0.3 ± 0.2 [0.2–0.6]	**3.2 ± 0.8** [2.3–3.8]	**10.0 ± 0.7** [9.2–10.6]
70	0.7 ± 0.4 [0.2–1.0]	**3.8 ± 0.9** [3.0–4.8]	**collapsed** (3/3)
80	1.3 ± 0.7 [0.6–1.9]	**6.9 ± 1.8** [5.0–8.6]	**collapsed** (3/3)

At 50 °C only 5 of 21 specimens reached the LOQ of 0.64 %; the 20–60 wt % values there cannot be ranked. At 60 °C 18 of 21 did, and at 70 °C all 15 intact specimens.

At 50 °C every composition passes (highest mean 1.3 % at 80 wt %). **At 60 °C the series splits:** 20–40 wt % pass at 0.6–1.5 %, while every specimen from 50 wt % up fails; leakage is nearly flat across 50–70 wt % (3.2–3.8 %) and doubles at 80 wt %. **At 70 °C nothing passes:** leakage climbs from 2.5 % to 10.0 % across 20–60 wt %, and 70–80 wt % collapse.

4.4.3 The retention limit X(T)

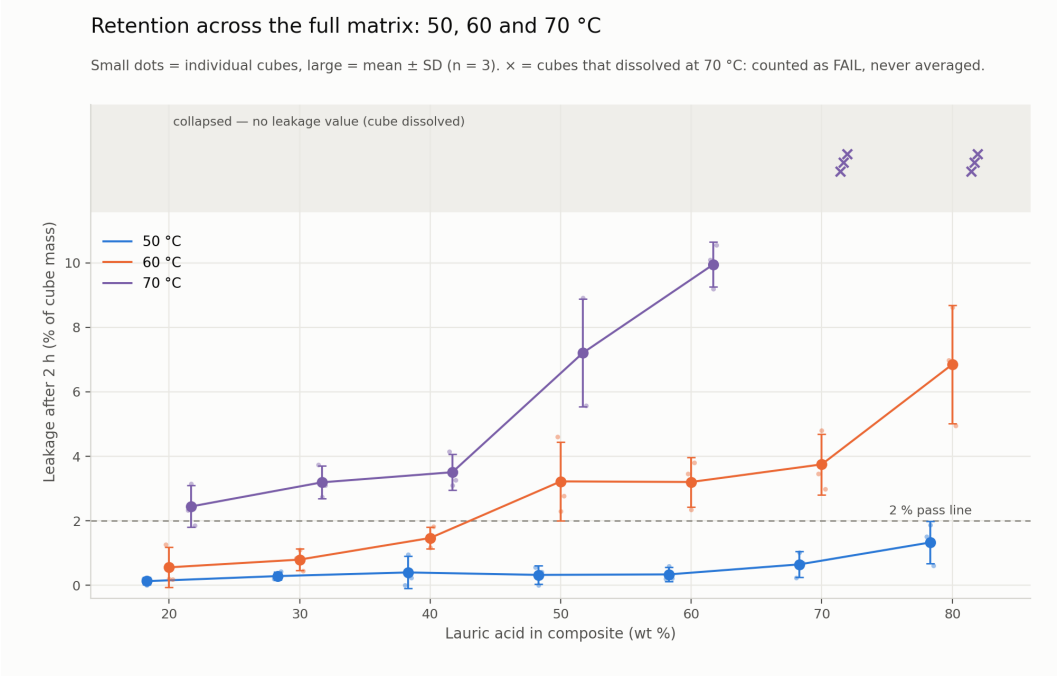


Figure 4.4. Leakage against lauric-acid loading at 50, 60 and 70 °C. Small points: individual specimens; large points: means ± SD (n = 3); crosses: specimens that collapsed at 70 °C, shown in a separate band because they have no leakage value. Dashed line: the 2 wt % criterion.

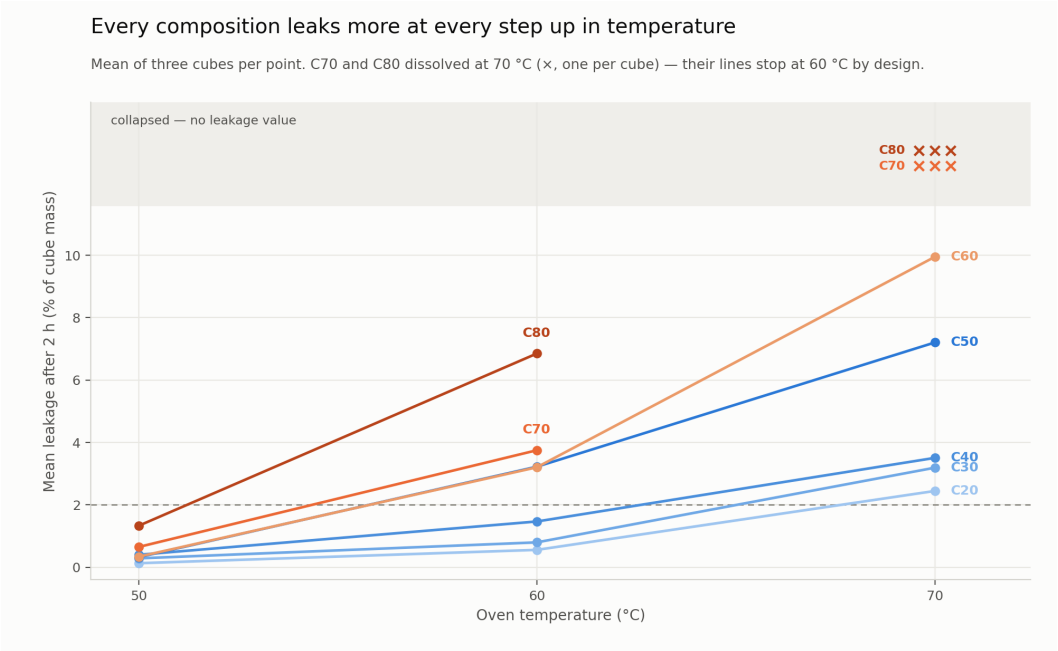


Figure 4.5. Mean leakage against temperature, one line per composition. Every composition leaks more at each step up in temperature.

Table 4.6. Retention limit X(T), interpolated as in §3.3.8. The bootstrap interval captures replicate scatter only and assumes linearity between adjacent compositions.

Temperature	X(T), wt % LA	Bracketing pair	Bootstrap 95 % CI
50 °C	**≥ 80**	top of tested range passes	not bracketed
60 °C	**43.0**	40 wt % pass (1.47 %) / 50 wt % fail (3.22 %)	41.2–46.4
70 °C	**< 20**	bottom of tested range fails (2.45 %)	not bracketed

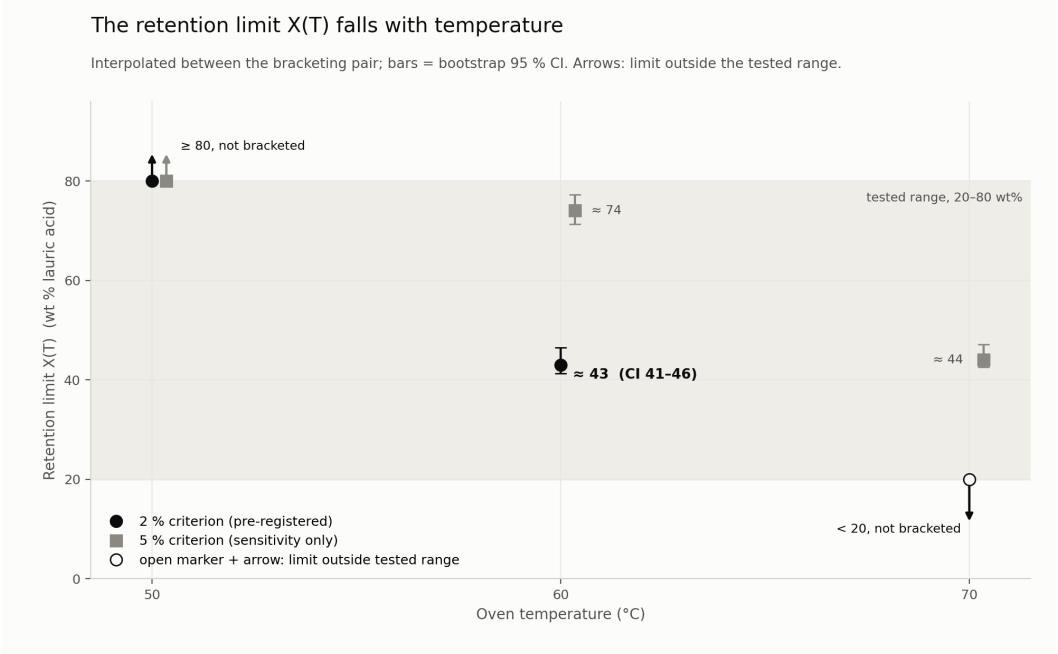


Figure 4.6. The primary result: retention limit X(T) against temperature. Filled circles, the pre-registered 2 % criterion; grey squares, a 5 % criterion (sensitivity check, not pre-registered). Arrows mark limits outside the 20–80 wt % range tested.

The retention limit falls from above 80 wt % at 50 °C to 43 wt % at 60 °C and below 20 wt % at 70 °C. Only X(60) is located; the others are bounds. The 2 % line is strict for a mostly-acid composite: assuming the escaped liquid is lauric acid (not analysed), every intact specimen kept at least ~83 % of its acid after two hours. A 5 % criterion moves X(60) to **74 wt %** (CI 71–77) and locates X(70) at **44 wt %** (CI 43–47); it shows how much the headline depends on the line and does not replace the pre-registered value.

4.4.4 Statistical tests

One-sided permutation tests (seed 20261003, 10^5 resamples); collapsed specimens rank above all measured values.

- **Temperature.** JT stratified by composition = 187.5 of a possible 189 (**$p < 10^{-5}$**); each step alone, 50 ⟶ 60 °C (U = 61.5 of 63) and 60 ⟶ 70 °C (U = 63 of 63), $p < 10^{-5}$. The mean rose at **7 of 7** compositions at each step (sign test p = 0.008).
- **Composition.** Leakage rises with loading at every temperature (JT p = 0.0004 at 50 °C, $p < 10^{-5}$ at 60 and 70 °C), and the trend persists within the 20–60 wt % specimens that stayed intact ($p < 10^{-5}$ at 70 °C, 2×10^{-5} at 60 °C), so it is not an artefact of the collapsed cells.
- **Collapse.** At 70 °C, 6 of 6 specimens at ≥ 70 wt % collapsed against 0 of 15 at ≤ 60 wt % (Fisher $p = 2 \times 10^{-5}$). This is **descriptive only**, since the split was read off the data; whether failure at 70 °C is a step or a steep ramp is not resolved.

4.4.5 Measurement validity

Mass balance. For intact specimens losing ≥ 0.03 g (n = 41), paper gain against specimen loss has slope **0.87** and **r = 0.976** (Figure 4.7). The paper recovered **84 %, 91 % and 89 %** of the loss at 50, 60 and 70 °C; the small unaccounted remainder (mean 0.004, 0.011 and 0.028 g) grows with temperature, consistent with evaporation or liquid running past the paper. The dish and sealed-control masses will separate the two [UPDATE once transcribed]. Either way, the primary figure measures liquid leaving the specimen, not paper moisture or handling.

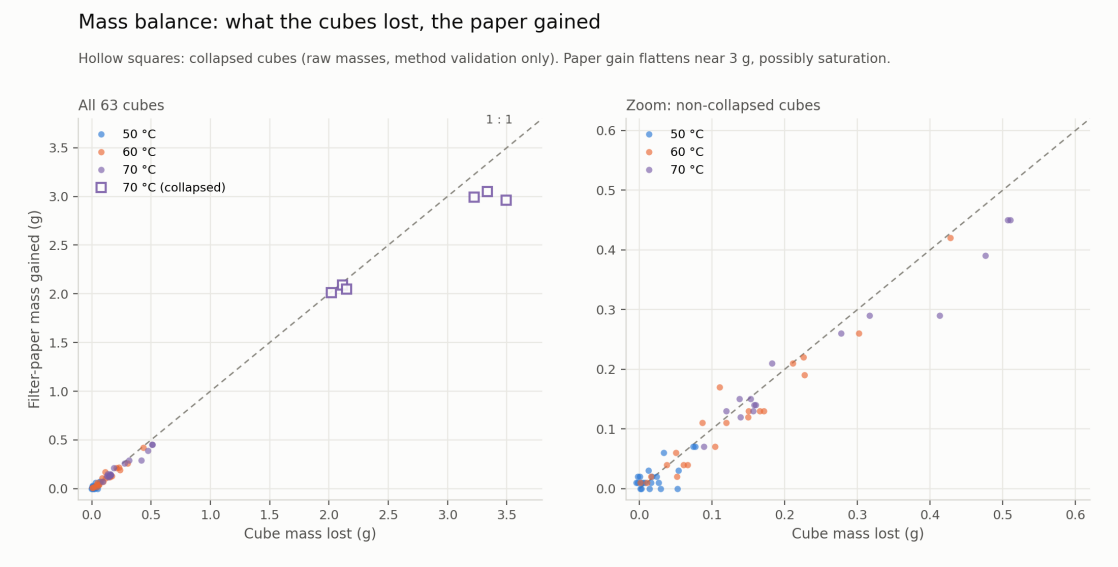


Figure 4.7. Paper gain against specimen loss for all 63 specimens (left) and intact specimens only (right), with the 1 : 1 line.

Run-to-run variation. No oven run read systematically high or low (within-cell permutation p = 0.62, 0.62, 0.69). Pooled within-cell SDs (0.38, 1.0 and 0.9 % at 50, 60 and 70 °C) are small against the differences between temperatures.

Collapsed specimens left 47–58 % (70 wt %) and 26–36 % (80 wt %) of their mass as residue, and their papers gained 2.0–3.1 g. These are recorded as "collapsed, leakage not separable" and used nowhere as leakage values.

4.4.6 Outcome against the pre-registered hypotheses

- **H1 (retention depends on loading): supported.** Pass and fail both occur; at 60 °C the series divides between 40 and 50 wt %.
- **H2 (service limit): not falsified.** Every composition leaks more at each temperature step. The 2 % crossing lies **between 50 and 60 °C for 50–80 wt %** and **between 60 and 70 °C for 20–40 wt %**; both intervals overlap the predicted 55–68 °C, but three temperatures cannot locate the crossing within them. The result is consistent with the prediction, not a test of it at that resolution.
- **H3–H6** depend on DSC, FTIR and SEM (§4.6–4.8) and, for H4's retention component, on cycling (§4.5). **H7** is reported in §4.9.

4.5 Thermal cycling stability (C40)

4.5.1 What was run

C40 was cycled as the highest loading passing at 60 °C. Logged hot legs ran 24–28 min and sum to **634 min (≈ 10.6 h)** over 25 cycles (the workbook's 609 min omits cycle 18, whose logged times give 25 min).

Three cool legs were short of the 25 min target (6 min after cycles 7 and 18, 4 min after cycle 22); whether the specimens were dull white before re-entry is [CONFIRM]. Whether the cycled specimens came from the original casting is [CONFIRM]; the baseline comparison is valid only for the same casting. The SOAK arm (2 × C40, 634 min at 60 °C) is [PENDING].

4.5.2 Result

Table 4.7. C40 after 25 melt–freeze cycles at 60 °C (≈ 10.6 h total hot time).

Specimen	m_0 (g)	m_1 (g)	Leakage %	Paper gain (g)	Dish gain (g)	Transfer %	Closure (g)	Appearance
CYC-1	5.04	4.22	16.3	0.45	0.18	12.5	0.19	intact, hollow at top
CYC-2	4.99	4.17	16.4	0.40	0.28	13.6	0.14	intact, hollow at top
CYC-3	4.31	3.64	15.5	0.39	0.27	15.3	0.01	intact, hollow at top
Mean ± SD			**16.1 ± 0.5**			13.8	0.11	0/3 collapsed
SOAK (n = 2)			**[PENDING]**					
Baseline, 2 h (n = 3)			**1.5 ± 0.3**					intact

Transfer % is not yet blank-corrected [enter BLANK-CYC masses].

The cycled specimens lost 16.1 ± 0.5 %, eleven times the two-hour baseline (1.47 ± 0.33 %). If the escaped liquid is lauric acid, that is ≈ 40 % of the acid present, against ≈ 4 % after two hours. None collapsed, but all three had a hollow in the upper face, consistent with liquid draining downward. The dishes gained 0.18–0.28 g; without them Transfer would have recovered about half the loss, with them 77–99 %. Two specimens exceed the 0.4-point Leakage–Transfer flag and are kept, as §3.3.7 requires.

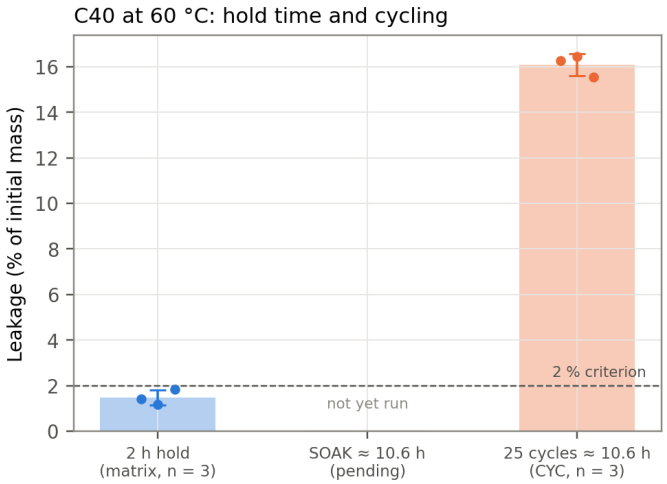


Figure 4.8. C40 at 60 °C: leakage after the 2 h matrix hold and after 25 melt–freeze cycles (≈ 10.6 h hot). Points: individual specimens; bars: mean ± SD. The soak arm is pending.

4.5.3 What the comparison can and cannot yet say

Hot time differs between the arms, so the increase cannot yet be attributed to cycling. Linear accumulation at the two-hour rate would give **≈ 7.8 %** over 10.6 h, half the observed loss, and hold-time curves for other composite classes flatten rather than rise linearly (Khalaf Alshammari et al., 2024). The **SOAK arm** decides [PENDING]:

- **SOAK ≈ CYC (within 1 pp):** the loss is caused by time at 60 °C, not by melting and freezing, and the two-hour X(T) is an early-time value.
- **SOAK < CYC by more than 1 pp:** repeated phase change loosens the network beyond hot time.
- **SOAK > baseline in either case:** retention at 60 °C is time-dependent, and X(T) must carry its hold time.

H4, retention component: not met by the literal criterion (≥ 5 % drift; leakage rose from 1.5 % to 16.1 %). Attribution awaits SOAK. The enthalpy, onset and supercooling components come from DSC (§4.6) and the new-band criterion from FTIR (§4.7). The tilted settling assay was not run (§3.5.2); the "hollow at top" is the small-scale version of the failure it was designed to detect.

4.6 Thermal properties by DSC

[PENDING — GIKI results.] Programme as requested (§3.7, Appendix C); [confirm the programme actually run from the facility report].

Table 4.8. DSC results (second heat unless stated).

Sample	Onset (°C)	Peak (°C)	ΔH (J/g)	Predicted ΔH = 191 × w (J/g)	Crystallisation onset (°C)	Supercooling (°C)
Raw RBW (PR 1798)				—		
Pure lauric acid				191		
C30				57		
C40 *(optimum, if submitted)*				76		
C50				96		
C70				134		
C40 after cycling *(if submitted)*				76		

Statements to make once the data are in:

1. **Wax melting point:** does DSC confirm the 70–73 °C arrest (§4.1) or the 77–79 °C literature value?
2. **H3:** ΔH against w(LA) with the zero-intercept 191 J/g line. A slope 4–7 % low is the expected offset (§1); H3 is falsified only by **non-linearity** beyond scatter.
3. **Hidden transitions** between the acid and wax melts.
4. **H4, supercooling:** composite against neat acid at matched mass and cooling rate.
5. **H4, enthalpy drift:** cycled against uncycled C40, criterion < 5 %.

4.7 Spectroscopic analysis (FTIR / ATR)

[PENDING — GIKI results.] Samples: pure LA, raw RBW, composites [30 / 50 / 70 or all seven], cycled C40.

Table 4.9. Carbonyl band positions (cm^{-1}).

Sample	Acid C=O (~1700)	Ester C=O (~1735)	Shift vs pure component	New bands?
Pure LA		—	—	—
Raw RBW	—		—	—
C30 / C50 / C70				
C40 cycled				

Statements to make: **H5**, whether the composite spectrum is a superposition (a carbonyl shift growing with composition would indicate acid–ester hydrogen bonding; check also the 2500–3300 cm^{-1} O–H region); **H4**, whether new bands appear after cycling, including any oxidation from the staggered oven residence at casting.

4.8 Microstructure (SEM)

[PENDING — run / not run.] Samples: raw RBW, one low- and one high-LA composite, uncycled and cycled C40. Statements to make: **H6**, network density and needle length (ImageJ, n ≥ 30) against pass/fail in Table 4.5; and whether the cycled network is coarser. If SEM was not obtained, H6 is reported as **untested**, not failed.

4.9 Field evaluation

Three days were logged in Al Khobar (15–17 October 2026) with four insulated-back mini panels: bare, RBW alone, C40 and C40 + graphite, each slab 14 × 7 × 1 cm (§3.6). Every reading is kept: 6,514 on Day 1 (from 13:30), 7,417 on Day 2 (from 11:47) and 12,725 on Day 3 (from 11:45, past sunset), with no gaps or failed reads. Modelled cloud never exceeded 7 %. No panel correction is applied (§6).

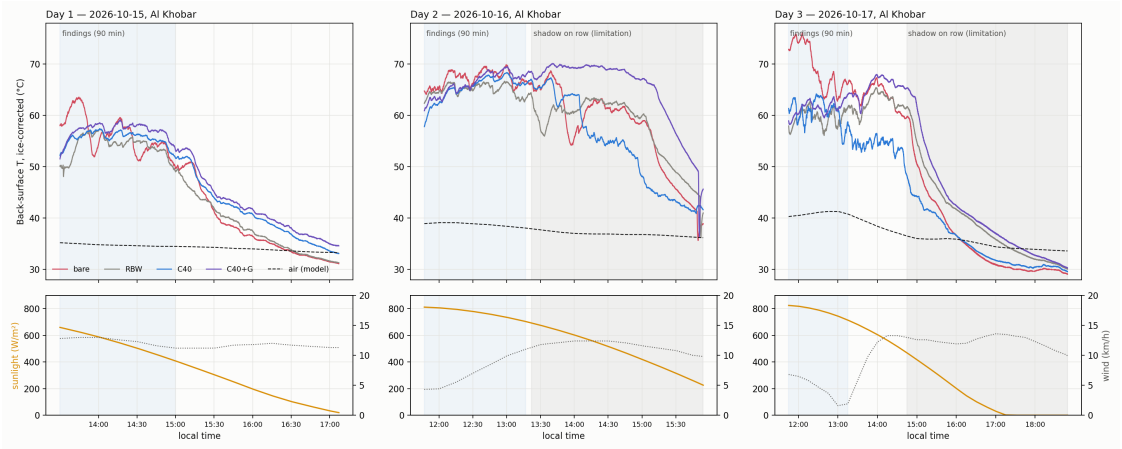


Figure 4.9. Back-surface temperature of the four stations on each day, ice-corrected, with modelled air temperature (dashed), sunlight and wind (Open-Meteo model data). Blue: the first 90 min (findings window, post hoc). Grey: the row in shadow (Day 2 from ≈ 95 min; Day 3 from ≈ 180 min). No reading is removed.

Table 4.10. Pre-registered main number: each slab station's daily peak subtracted from the bare station's (positive = cooler than bare), and slab mass over the three days outdoors (1 g scale).

Station	Day 1	Day 2	Day 3	Mean ± SD	95 % CI (n = 3)	Mass before → after	Mass loss
Bare peak (°C)	63.6	69.9	76.1	—	—	—	—

Station	Day 1	Day 2	Day 3	Mean ± SD	95 % CI (n = 3)	Mass before → after	Mass loss
B — RBW alone	+6.2	+3.2	+10.7	**+6.7 ± 3.7**	−2.6 to +16.0	98 → 98 g	**0 %**
C — C40	+6.1	+1.6	+11.9	**+6.5 ± 5.2**	−6.3 to +19.4	96 → 94 g	**2.1 %** (±≈ 1 %)
D — C40 + graphite	+4.5	−0.2	+8.1	**+4.1 ± 4.2**	−6.2 to +14.5	100 → 89 g + fragments	**11 %**
C − B (latent over mass)	−0.1	−1.7	+1.2	−0.2 ± 1.4	−3.8 to +3.4		
D − C (graphite)	−1.7	−1.8	−3.8	−2.4 ± 1.2	−5.4 to +0.6		

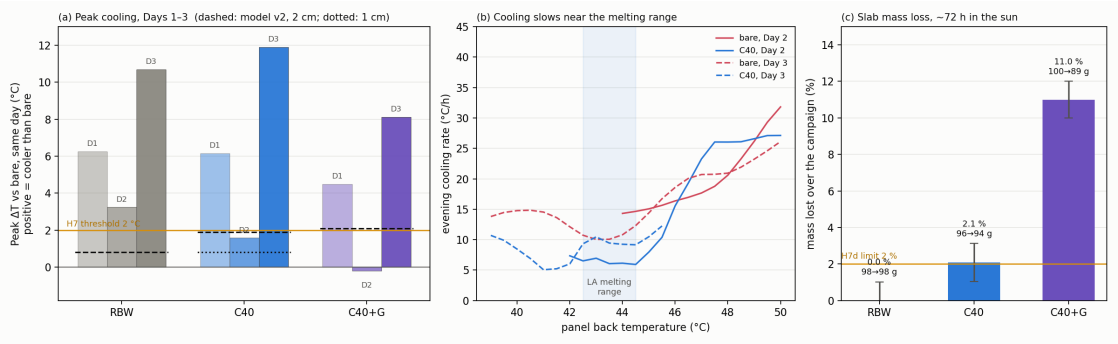


Figure 4.10. (a) Daily peak ΔT against bare for each slab (D1–D3), with the pre-registered v2 prediction for 2 cm slabs (dashed) and the same model at the as-built 1 cm (dotted, C40); orange, the 2 °C H7 threshold. (b) Evening cooling rate against module temperature after the row loses sun, bare and C40, Days 2 (solid) and 3 (dashed); blue band, lauric acid's melting range (post hoc). (c) Slab mass loss; bars ±1 g; orange, the 2 % H7d limit.

4.9.1 Peak cooling (H7)

The C40 slab lowered the daily peak by **6.5 °C** on average, more on hotter days: 1.6 °C on Day 2, 6.1 °C on Day 1 and 11.9 °C on Day 3, when bare reached 76.1 °C. The bare panel had the highest peak every day except Day 2, when the graphite station exceeded it by 0.2 °C.

H7: the mean exceeds the pre-registered 2 °C threshold, but the effect is not statistically established. On the field pre-registration's criterion (falsified if mean ΔT < 2 °C) H7 is met. On the stricter condition of §1.2 it is not established: the design had no crossover, so no crossover correction is possible, and with three days and one module per treatment the 95 % CI (−6.3 to +19.4 °C) includes zero. The direction was the same on all three days. The measured mean is more than three times the v2 prediction for a 2 cm slab (+1.9 °C) and eight times the +0.8 °C the same model gives for 1 cm.

At matched sunlight (≥ 600 W/m², post hoc) the slab panels averaged 6.0 °C (C40), 5.4 °C (RBW) and 3.5 °C (C40 + graphite) cooler than bare. In the evening the order reversed as the slabs released stored heat (Day 3, last 30 min: bare 29.7 °C, RBW 30.9 °C, C40 30.3 °C, C40 + graphite 31.1 °C).

4.9.2 Latent heat over thermal mass (H7a)

Across the three days, wax alone cooled the peak as much as C40 (C − B = −0.2 ± 1.4 °C), so H7a, as pre-registered, is not supported. On the hottest day, however, the acid's latent heat is visible: from 80 min until the row lost sun, C40 held near 55 °C while wax alone climbed to 62–64 °C.

Table 4.11. Day 3, ice-corrected module rear temperature (°C, 2 min means), minutes after 11:45. Negative C − B means C40 cooler than wax alone.

Minute	40	75	82	100	120	150	175
Bare	68.8	63.1	65.8	64.7	64.4	64.0	59.8
B — RBW alone	59.7	57.8	60.2	61.2	62.2	63.6	61.7
C — C40	63.1	61.0	54.6	54.6	54.6	54.7	52.0
C − B	+3.4	+3.2	−5.6	−6.6	−7.6	−8.9	−9.7

- **Day 3, 13:05–14:45.** For the first 78 min C40 ran 1–5 °C *warmer* than wax alone. At ≈ 80 min its reading fell ≈ 6 °C in about four minutes while the other three kept heating, then held almost flat at 54.6–54.7 °C for over an hour, **≈ 8 °C cooler than wax alone and ≈ 10 °C cooler than bare** (95–180 min means: C40 54.1 °C, RBW 62.4 °C, bare 63.6 °C) until the row lost sun (Table 4.11). The row was checked every 30 min and the C40 station was not shaded, and after sunset the C40 and wax stations read within 0.54 °C of bare, so panel bias cannot account for an 8 °C gap. A module temperature held flat while its neighbours keep heating is the signature of latent absorption at the interface the sensor reads, and the slab's latent capacity, 96 g × ≈ 76 J/g ≈ 7 kJ, is of the order needed if most of the acid melted in this window. The abruptness of the initial fall suggests that the softening slab also settled into closer contact with the module and sensor (a light smudge of composite was later found around the sensor); contact and latent heat cannot be separated with these data. On Days 1 and 2 no such margin appeared (C40 0.1 and 1.7 °C warmer than wax at peak).
- **Evening freezing (Figure 4.10b, post hoc).** On Day 2, C40 cooled at **5.9 and 4.5 °C/h at 44 and 42 °C**, against 14.1 and 23.7 °C/h for bare at the same temperatures. On Day 3 the contrast was weaker (C40 8.7 and 6.1; bare 10.2 and 12.4 °C/h), and RBW and C40 + graphite also slowed near 42 °C. This is the freezing arrest H4 predicts at bulk scale.

H7c (melt by day, freeze by night, from outer-face sensors) was not tested: no outer-face sensors were fitted. The evening cooling rates are indirect evidence only.

4.9.3 Graphite (H7b)

Graphite did not help. C40 + graphite was hotter than C40 at peak every day (by 1.7, 1.8 and 3.8 °C; −2.4 ± 1.2 °C, CI −5.4 to +0.6), the opposite of H7b and of the model's negligible difference. By H7b's own criterion (CI includes 0) it is falsified. The slab partly collapsed and flowed on Day 2 after ≈ 105 min above 68 °C and broke apart by the end of Day 3 (Figure 4.11c). Lost contact with the module, rather than conductivity, is the likely reason it cooled less. Whether graphite itself weakens the network is untested: the side-by-side oven hold of C40 and C40 + graphite has not been run [PENDING — C40 vs C40 + graphite oven test at 60 and 70 °C, §3.6.2].

4.9.4 Retention in service (H7d)

Over three days outdoors, with module peaks of 60–76 °C, wax alone lost nothing measurable (98 → 98 g), C40 lost **2.1 %** (96 → 94 g) and C40 + graphite **11 %** (100 → 89 g plus fragments) (Figure 4.11). **C40 sits at the 2 % limit and cannot be separated from it on a 1 g scale; C40 + graphite clearly fails H7d.** Because wax alone lost nothing, C40's loss is attributable to the acid. In the oven C40 lost 1.47 % in 2 h at 60 °C, 3.51 % in 2 h at 70 °C and 16.1 % over 25 cycles; the field slab lost far less over far longer, but it was heated from one face, cooled nightly and held against a board rather than on absorbent paper, so the figures are not directly comparable. It is the first multi-day retention measurement for this composite in service.

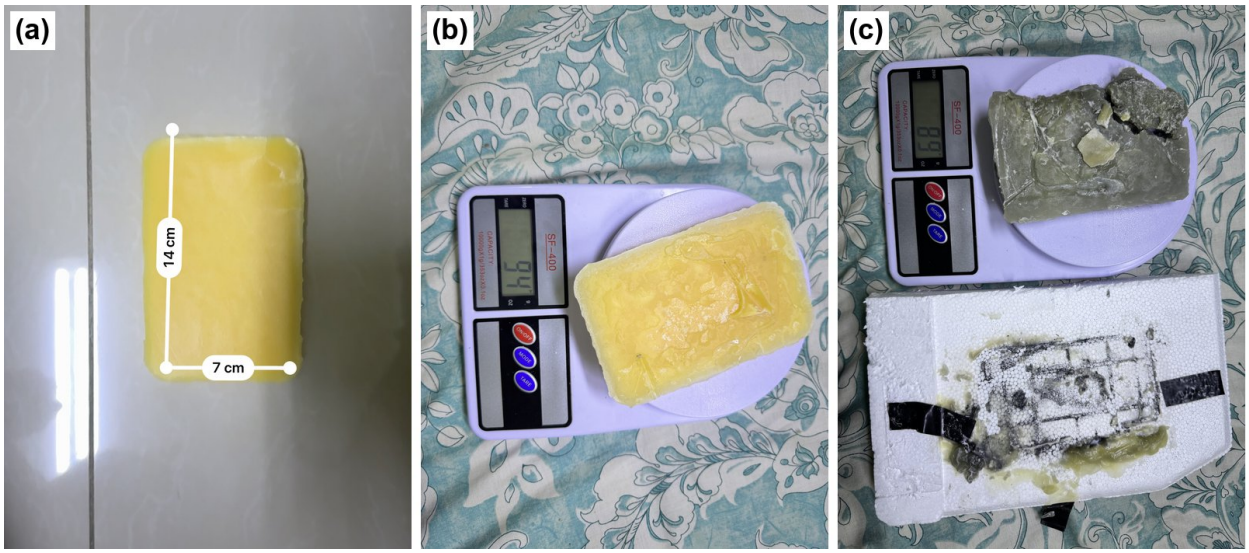


Figure 4.11. Slabs after the campaign. (a) A cast slab, 14 × 7 cm, 1 cm thick. (b) The C40 slab on the scale at 94 g (96 g before), intact. (c) The C40 + graphite slab at 89 g (100 g before), broken, above its board, which shows a graphite print where the slab sat and composite pooled along the lower edge. Photographs taken by the authors.

4.9.5 What the field result can and cannot support

It supports: a 1 cm rice-bran-wax slab, with or without lauric acid, lowered the peak of an insulated-back mini module by several degrees in Al Khobar's October sun on each of three days; on the hottest day the acid held the C40 module ≈ 8 °C below the wax-only module for ≈ 1.5 h; C40 kept its mass to ≈ 2 % over three days outdoors; graphite as dispersed here harmed both cooling and stability. It does not support a statistically established effect, a latent-heat margin over thermal mass on every day, or transfer of the absolute numbers to full-size modules (§6).

4.10 Panel-scale thermal model: analysis

The model of §3.9.6 is assessed here against the field result, then analysed to show which properties its prediction depends on. The analysis sweeps are post hoc and do not replace the prediction.

4.10.1 Version 2 against the field

For the insulated-back design actually run, v2 predicted C40 **+1.9 °C** cooler than bare at peak (2 cm), wax alone +0.8 °C, and graphite no detectable change. The field gave **+6.5 °C** (C40) and **+6.7 °C** (wax) with 1 cm slabs, for which the model gives only +0.8 °C. **The model got the sign right and the size wrong by a factor of three or more, and its predicted latent margin of C40 over wax (≈ 1.1 °C) was not seen on average.** Three model features plausibly explain the shortfall: its module heat capacity (9000 J/m²·K) and convection correlation are for full-size modules, whereas a 1 W mini panel has little thermal mass of its own, so a slab of comparable mass dominates; it ran on a climate 12–18 °C cooler than Al Khobar's; and its properties are placeholders. The model predicted that composite and graphite slabs would melt partly by day and re-solidify by night (peak liquid fraction 0.25, 0.00 every dawn); the evening cooling arrest near 43 °C (§4.9.2) is consistent with that.

4.10.2 Version 1 and why a layer can heat a module

The first prediction (open rack, convection only, ΔH 95 J/g) gave **−5.3 °C**: the layer made the module hotter (Figure 4.12). Its composite peaked at only **48.4 °C**, used **28.8 kJ** of **171 kJ** latent capacity (liquid fraction 0.17) and re-solidified every night. The reason is the back path. On an open rack, the 2 cm layer

adds 0.100 m²K/W in series with a 0.125 m²K/W convective film, cutting steady rear conductance by **44 %**, while its storage acts only near the peak. Against insulation the rear is already resistive, so the slab adds little resistance and its storage dominates (§3.9.6). **Mounting decides the sign,** which is why the field test used insulated backs, and why an open-rack PCM unit made modules hotter in Soares et al. (2020).

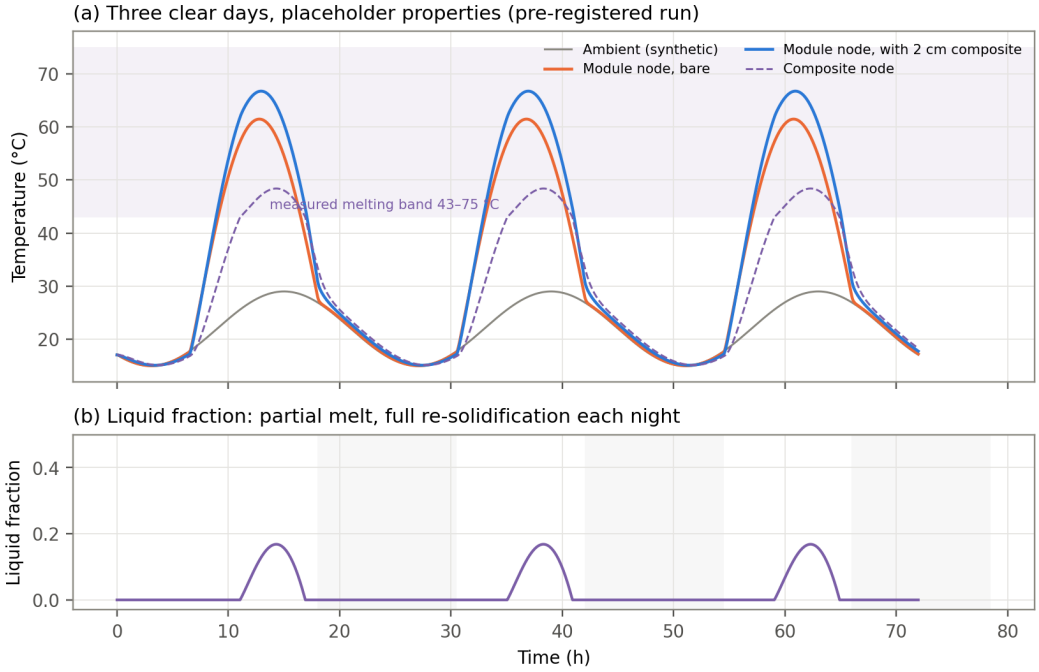


Figure 4.12. Model v1 (open rack, pre-registered first) over three synthetic clear October days. (a) Bare module, module with a 2 cm composite layer, and composite node, with the measured melting band shaded. (b) Composite liquid fraction; night shaded.

Numerical check. Halving the timestep from 30 s to 1 s changes the peak ΔT by 0.001 °C. The real approximation is lumping: the layer Biot number is 0.80, so the panel-side face would run above the lumped temperature and melt somewhat more, making the lumped result mildly pessimistic about melting.

4.10.3 Sensitivity to conductivity, latent heat and thickness (v1, post hoc)

On the open rack, conductivity dominates (Figure 4.13): peak ΔT rises from −9.7 °C at k = 0.1 W/m·K to **break-even at $k \approx 0.8$ W/m·K** and +1.7 °C at 6 W/m·K, never reaching 2 °C at 2 cm and 95 J/g. At k = 0.2 W/m·K every combination of latent heat (50–191 J/g) and thickness (0.5–5 cm) warms the module; at k = 1.0 W/m·K the 2 °C target needs **ΔH ≥ 150 J/g at 3 cm or 191 J/g at 2 cm**, essentially neat lauric acid (Figure 4.14). On the insulated back (v2), by contrast, k from 0.2 to 6 W/m·K moves peak ΔT only between +1.6 and +2.2 °C.

4.10.4 Benchmark against a sharp-melting paraffin

With RT44-like properties (43–44 °C, 250 J/g, 800 kg/m³, 2 cm), the same open-rack model also warms the module at k = 0.2 W/m·K (−3.75 °C) but cools it by **+9.9 °C** at k = 1.0 W/m·K, melting fully, where the composite gives +0.4 °C. **The composite's 32 °C-wide melting band, not its onset, is its main thermal handicap:** the latent heat arrives too slowly and too high. The large RT44 reduction of Durez et al. (2023) is not reproduced at k = 0.2 W/m·K, pointing to differences in climate and in how the back boundary and layer resistance are treated.

[PENDING — re-run v2 with measured ΔH (DSC) and k (hot-wire), and on Al Khobar weather, as a post-hoc check.]

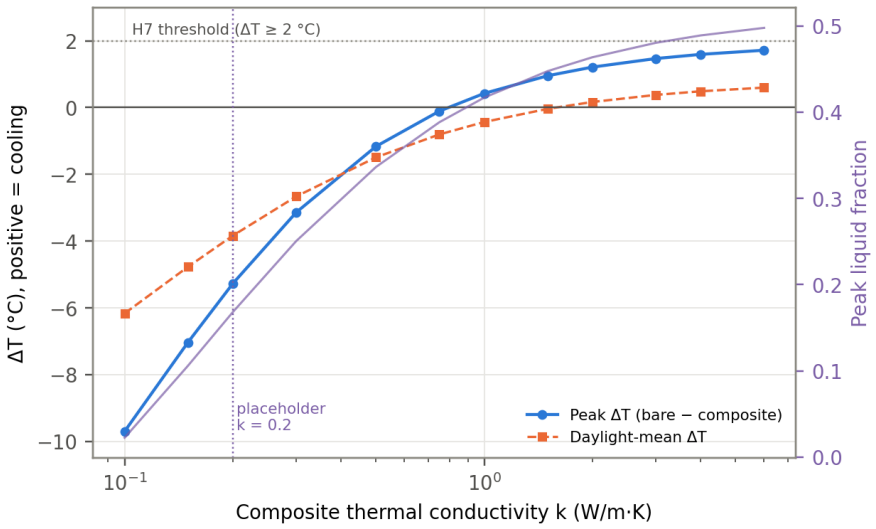


Figure 4.13. Model v1 (open rack), post-hoc sensitivity to composite conductivity at 2 cm and 95 J/g: peak and daylight-mean ΔT (positive = cooling, left axis) and peak liquid fraction (right axis). Dotted line: the pre-registered placeholder.

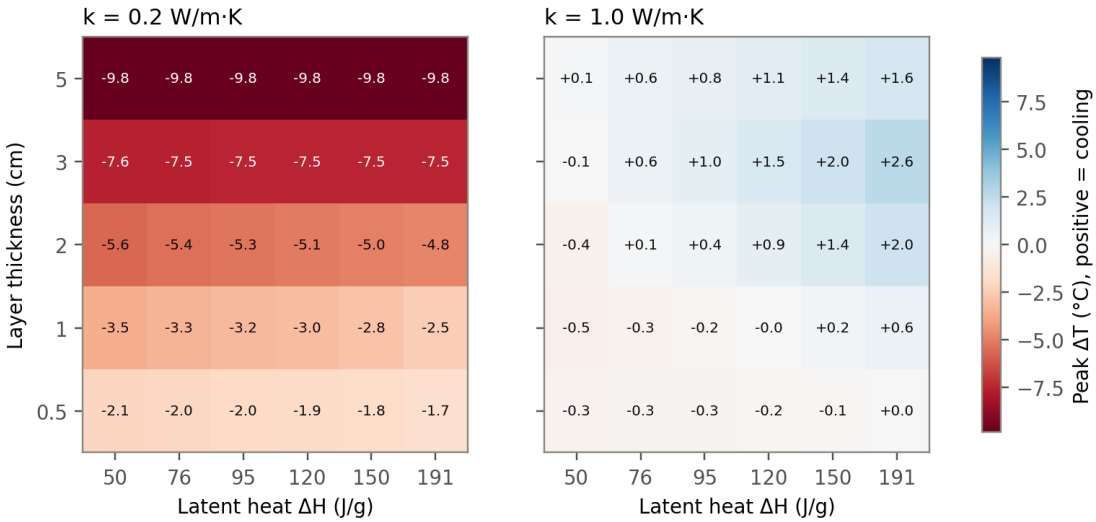


Figure 4.14. Model v1 (open rack), post-hoc sensitivity of peak ΔT to latent heat and layer thickness at k = 0.2 W/m·K (left) and 1.0 W/m·K (right). Blue, cooling; red, warming.

5. Discussion

5.1 What the retention surface shows

Across a 20 °C span the network goes from holding four times its own mass of liquid acid to failing at the lowest loading tested (X(50) ≥ 80, X(60) = 43, X(70) < 20 wt %; §4.4.3). H1 is supported and H2 not falsified. The predicted 55–68 °C service limit is consistent with the data but only loosely tested: the data contain the predicted window and do not contradict it, but do not place the limit to better than 10 °C.

The 50 °C result stands on its own. With the acid fully molten, twenty parts of wax held eighty parts of liquid and lost 1.3 % in two hours. Rice bran wax structures triglyceride oils from 0.5 wt % (Wijarnprecha et al., 2018); the present data show the same network forming in a **free fatty acid** melt, which, to our knowledge, had not been shown. The question of §2.3, whether a wax network can hold a fatty-acid melt at all, is answered yes at 50 °C.

5.2 Reconciling the thermodynamic and mechanical measurements

The composite's liquidus is near **75 °C** (§4.2), yet from 50 wt % up it fails by 60 °C, while much of its wax should still be solid. **The gap is a finding, not a discrepancy**: the melting curve detects whether *any* solid remains, the leak test whether the solid forms a **continuous** network. As crystals dissolve, the remainder can fall below the fraction needed to stay connected across the specimen long before the last crystal disappears, and liquid then drains past solid that is still present. Retention should therefore fail at a **critical solid fraction**, not at the liquidus.

The composition series fits this qualitatively. More acid lowers the limit in two ways at once: less wax to begin with, and more solvent to dissolve it. At 70 °C the 50–60 wt % specimens stayed intact and leaked 7–10 %, while 70–80 wt % collapsed; whether this is a sharp percolation threshold or a steep ramp is unresolved (§4.4.4).

The ideal-mixing model underestimates how much wax stays solid. It predicted a liquidus of ≈ 67 °C at 50 wt %, yet the 50 and 60 wt % specimens stayed intact at 70 °C, in line with the ≈ 75 °C water-bath liquidus. For chemically similar fatty acids, the same Schröder approach predicts measured eutectics closely (Zhou et al., 2026); RBW is mostly long-chain esters, unlike the acid dissolving it, so a departure from ideality is expected. **A critical solid fraction would be more transferable than a temperature:** with the wax solubility curve from the DSC series (§4.6 [PENDING]), the solid fraction in each cell of Table 4.5 could be computed, and a single failure value would predict X(T) for any loading.

5.3 Time dependence: the two-hour limit is not the whole answer

After 25 cycles C40 lost **16.1 %**, eleven times its two-hour loss (§4.5). Whatever the cause, **retention in this network is strongly time-dependent**, and every X(T) here is a two-hour figure. This sets it apart from other composite classes: hexadecane / LDPE composites lost 0.10–3.64 % in the first hour and then levelled off within about 14 h (Khalaf Alshammari et al., 2024), and paraffin / expanded-graphite composites held below 4 wt % over ten 1 h holds at 60 °C (An et al., 2022). A 16 % loss after ten hours is of the order Wadee et al. (2025) report for coated PCM granules only after **500 h** at 50 °C (11.4–14.1 %). The mechanism of §5.2 explains why: a porous carrier's pores do not change with time, while a dissolving network can keep coarsening and losing connectivity.

[PENDING — SOAK.] Keep the paragraph that matches the result:

- *If SOAK ≈ CYC:* the loss is caused by sustained time at 60 °C; the network survives freezing and remelting, but not hours at temperature.
- *If SOAK < CYC by more than 1 pp:* each freeze–melt cycle disturbs the network beyond time alone, plausibly by recrystallisation into fewer, larger crystals. SEM on the cycled specimen (§4.8) is the direct test.

The "hollow at top" on all three cycled specimens points to **gravity-driven drainage**, the small-scale version of the settling failure the untested tilted assay was designed to detect (§3.5.2).

5.4 Is the acid–wax system a physical mixture? (H5)

[PENDING — FTIR.] The thermal data already favour H5: the liquidus lies *above* the ideal-mixing prediction, so the wax is less soluble than ideal, the opposite of what strong acid–ester hydrogen bonding would produce. (The earlier contrary indication, $\gamma \approx 0.05$, came from the withdrawn run, §6.)

- *If no carbonyl shift is seen:* H5 survives; retention is governed by network mechanics alone (§5.2).
- *If a reproducible shift is seen, especially one growing with acid fraction:* H5 is falsified, and the interaction must be reconciled with the low solubility, for example as one too weak to raise solubility but visible spectroscopically.

5.5 Implications for deployment

The application constraint is a rear temperature of **55–60 °C** on clear days in northern Pakistan, and well above that behind roof-integrated modules (§1; the field bare panel reached 64–76 °C). Against it:

1. **At 60 °C only loadings ≤ 40 wt % meet the criterion, and then only for two hours** (§4.4, §4.5).
2. **Containment costs more than half the storage.** At 40 wt %, the pre-registered rule gives ≈ 76 J/g against 191 J/g for neat acid [DSC value].
3. **On a roof-integrated module, mass matters more than conductivity or latent heat.** In the field, 1 cm slabs cooled the peak by ≈ 6.5 °C with or without the acid (§4.9), far more than model v2 predicted (+0.8 °C for 1 cm), and the model shows why the mounting matters: on an open rack the same layer insulates and warms the module (v1, −5.3 °C; Soares et al., 2020, measured the same effect), while against insulation storage dominates and conductivity barely matters (§4.10). Graphite at ≈ 3.8 wt % made the slab warmer and unstable rather than better. Where conductivity does matter, on an open rack, break-even needs $k \approx 0.8$ W/m·K, and 10–15 wt % expanded graphite in a fatty-acid eutectic raised k only 2.4–2.6-fold (Wang et al., 2022), to ≈ 0.5 W/m·K from the 0.2 W/m·K placeholder. Pressed expanded graphite reaches ≈ 16 W/m·K (Wang et al., 2024), but then holds the PCM by capillarity, making it a porous-carrier composite. In the same model a sharp-melting paraffin cools ≈ 25 times more than the composite at $k = 1$ W/m·K: the **32 °C-wide melting band**, more than the onset, limits the composite thermally.
4. **At 50 °C the material works well.** A shaded or ventilated installation, a milder climate or a non-PV thermal buffer whose peak stays near 50 °C is within the measured envelope.

What the field test adds. Most of the measured cooling is thermal mass, which the cheaper, non-leaking wax delivers alone (6.7 °C against 6.5 °C for C40). The acid's latent heat showed clearly on the hottest day, when the C40 module held near 55 °C for ≈ 1.5 h while the wax-only module rose to 62–64 °C, and as slow evening cooling near 43 °C, but not on the two milder days; its cost was ≈ 2 % of slab mass in three days. Panel bias was not corrected, but after sunset the RBW and C40 panels converged to within ≈ 0.5 °C of bare, small against the ≈ 6.5 °C effect (§6). **A wax slab is a credible passive peak-shaver for roof-integrated modules; the acid is worth adding only if its latent margin can be shown across many days with the slab contained.**

What can be claimed: an agricultural by-product forms a network that holds molten lauric acid at high loading at 50 °C, its retention limit across the PV range is measured with its time dependence attached, and on an insulated-back module a 1 cm slab lowered the peak by several degrees on each of three days. What cannot: that the composite as formulated is a deployable, long-life PV coolant. That would need a loading at or below about 40 wt %, a container that tolerates slow drainage, and multi-week evidence on full-size modules.

5.6 Comparison with porous-carrier stabilisation

The two mechanisms of §2.2 should differ in the **shape** of their retention curves. Capillary retention in a fixed pore geometry should be nearly flat with temperature above the melt and level off with time. A dissolving network should lose retention progressively with temperature, keep losing it with time, and end in collapse. **The present data show all three features of the gelled case:** X(T) falls steeply with temperature, loss grows with time, and the highest loadings collapse at 70 °C. By contrast, paraffin in an expanded-graphite / copolymer matrix stayed below 4 wt % over ten 1 h holds at 60 °C (An et al., 2022), and pressed expanded graphite held 80.8 wt % paraffin without leakage (Wang et al., 2024). No porous-carrier arm was run under the same protocol, so the contrast is drawn against literature; measuring it side by side is the most informative next experiment (§8).

6. Limitations

- **Material grade is supplier-stated.** The wax melted ≈ 5 °C below the literature range (§4.1), and gelation depends on ester composition; results describe batch PR 1798 until DSC resolves the grade.
- **A systematic error was found and corrected.** An early, incompletely mixed composite run gave internally consistent data, including a 10 min plateau at 48 °C, and survived four confound checks (melt transparency, blend provenance, a 24 h anneal, a solid network before heating). A service limit of ≈ 48 °C and a favourable interaction ($\gamma \approx 0.05$) were inferred from it. Only a changed preparation exposed the fault; the properly mixed composite melted to 75 °C, and the 48 °C result and everything derived from it were withdrawn. It is the reason the composition series used recorded masses and a fixed preparation (§3.2).
- **Instrument resolution.** A 1 °C thermometer for melting points; a 0.01 g balance on 4–5 g specimens (LOQ 0.64 %), so 20–60 wt % cannot be ranked at 50 °C (§4.4.2). Visual "fully melted" endpoints depend on heating rate (§4.1) and are not used.
- **Preliminary composite runs.** Masses were not recorded; the 75 °C liquidus rests on one properly mixed run (§4.2).
- **Specimen geometry.** 4.0–5.4 g pentagonal prisms, not the nominal 7 g; leakage percentages are specific to this surface-to-volume ratio.
- **Design resolution.** X(T) is located at one temperature only, the 60–70 wt % collapse boundary at 70 °C is unresolved, and the X(60) interval reflects n = 3 scatter and assumes linearity between compositions.
- **Hold time.** All X(T) values are two-hour values; leakage was measured at only 2 h and ≈ 10.6 h, so the shape of the leakage–time curve is unknown [update once SOAK reports].
- **Number of cycles.** Twenty-five cycles (≈ 10.6 h hot) is early-life behaviour against 500–700-cycle studies and 500 h exposures (Wadee et al., 2025; Du et al., 2021, as cited in Zhou et al., 2026).
- **Supporting masses and second channel.** Dish, sealed-control and late-run blank masses are not yet analysed [TRANSCRIBE]; papers were not photographed, so leakage rests on gravimetry alone.
- **Handling after the oven.** Papers came straight from the box (corrected by blanks), no desiccator was used, and specimen temperature was not measured before weighing; a specimen not fully cooled reads falsely low, so a small error of this kind cannot be excluded.
- **Pour order.** Compositions were poured 20 ⟶ 80 wt %, so later pours spent up to ~15 min longer molten after a common > 1 h soak. The effect is argued to be negligible but is aliased with the composition trend [reverse-order recast: run / not run].
- **Identity of the escaped liquid.** Treating it as lauric acid is an assumption; at 60–70 °C it carries dissolved wax.
- **Scale and orientation.** Specimens were small and horizontal; gravitational separation in a tilted panel was not tested (§3.5.2). DSC supercooling does not transfer to panel scale (§3.7).
- **Field design.** Three days with one module per treatment, so every 95 % CI on the daily peak difference includes zero, and no crossover correction was possible. Whole stations were rotated, so panel and sensor bias stay tied to treatment, and the bare calibration run's log was not available. After sunset the slab stations read at most **0.14 °C (RBW), 0.54 °C (C40) and 1.26 °C (C40 + graphite)** warmer than bare. Because heat still leaving the slabs adds to that gap, it bounds a bias that would make a slab panel read too warm (correcting for it would only increase the measured cooling), but it cannot exclude a bias of the other sign or one that appears only in sunlight.
- **Field measurements.** Sensors were calibrated at the ice point only, with span differences at panel temperature uncorrected (§3.8). They sat between module and slab, so contact between sensor, module and a softening slab may change as the slab melts and settles (§4.9.2); earlier records placed them below the slab edge and were corrected (§3.6.1). Weather is model data. The slabs had no container and were held between module and board, so their mass change combines leakage, evaporation and handling loss; mass was read on a 1 g scale, so C40's 2.1 % cannot be separated from the 2 % criterion. Slabs were 1 cm, not the predicted 2 cm, and mini panels are not full-size modules. The findings window, matched-sunlight comparison and evening cooling rates were added after the

data were seen. H7c was not tested.

- **External instruments.** DSC, FTIR and SEM were run by facility staff [PENDING results].
- **Thermal model.** ΔH, density, heat capacity and conductivity are placeholders; the model is lumped (layer Biot number 0.8), runs on a synthetic climate, and uses an order-of-magnitude module heat capacity. It is a pre-registered statement of expected behaviour, not a quantitative forecast, and it underestimated the field cooling by a factor of three or more.
- **Sample sizes.** Lauric acid melting n = 2; rice bran wax n = 3 (one superseded); properly mixed composite n = 1; retention matrix n = 3 per cell (63); cycling n = 3 cycled and n = 2 soaked ; field 3 days, one module per treatment.

7. Conclusions

1. **Rice bran wax forms a network that holds molten lauric acid:** at 50 °C every composite from 20 to 80 wt % lost ≤ 1.3 % in two hours (X(50) ≥ 80 wt %), to our knowledge the first such measurement for a natural-wax network holding a free fatty acid.
2. **The retention limit falls steeply across the PV range:** X(60) = 43 wt % (95 % CI 41–46) and X(70) < 20 wt %, with 70–80 wt % collapsing at 70 °C; leakage rose with temperature and loading ($p < 10^{-5}$). H1 is supported and H2 not falsified.
3. **Retention fails mechanically, well below the ≈ 75 °C liquidus,** consistent with loss of network connectivity rather than complete dissolution of the wax.
4. **There is no useful eutectic:** the 43–44 °C onset matches pure lauric acid; the wax contributes containment only.
5. **Retention is strongly time-dependent:** over 25 cycles (≈ 10.6 h at 60 °C) C40 lost 16.1 ± 0.5 %, eleven times its two-hour loss. [PENDING — SOAK: cycling or time.]
6. [PENDING — DSC: H3, latent heat against acid fraction; the wax melting point.]
   [PENDING — FTIR: H5.] [PENDING — SEM: H6.]
7. **Mounting decides whether a PCM layer cools or heats a module:** the pre-registered model predicted warming on an open rack (−5.3 °C) and cooling against insulation (+1.9 °C), and the composite's 32 °C-wide melting band is its main thermal handicap.
8. **In the field, a 1 cm slab lowered the peak of an insulated-back mini module by several degrees, mostly through thermal mass:** C40 by 6.5 ± 5.2 °C and wax alone by 6.7 ± 3.7 °C over three days in Al Khobar, above the 2 °C threshold on the mean but not statistically established (95 % CI includes zero). The acid's latent heat held C40 ≈ 8 °C below wax alone for ≈ 1.5 h on the hottest day but gave no margin on the two milder days (H7a, as pre-registered, not supported), graphite made the slab warmer and unstable (H7b falsified), and over three days C40 lost 2.1 %, wax 0 % and C40 + graphite 11 % (H7d: C40 at the limit, graphite fails).

Together, the results put a quantified ceiling on a passive-cooling material made from an agricultural residue: it works at 50 °C, is marginal at 60 °C and fails at 70 °C, and its two-hour figures overstate retention under sustained heat. Behind a roof-integrated module, even the wax alone shaved several degrees off the daily peak.

8. Future work

- **A hold-time series (highest priority).** Fresh C40 at 60 °C withdrawn at 0.5, 1, 2, 4 and 10 h, tying the matrix and soak results into one leakage–time curve; then at 50 °C and across loadings, so X(T) carries its hold time.

- **Locate the limit.** Test at 55 and 65 °C and at 45 and 65 wt %, plus a composition between 60 and 70 wt % at 70 °C to tell a step from a ramp.
- **Critical solid fraction.** Build the phase diagram from the DSC series and test whether failure occurs at one solid fraction (§5.2).
- **A porous-carrier arm** (lauric acid in expanded perlite or diatomite) under the identical protocol, to measure the curve-shape difference of §5.6.
- **C40 against C40 + graphite in the oven** at 60 and 70 °C, and conductivity by transient hot-wire, to tell whether graphite weakens the network; a hybrid (acid in a graphite or porous carrier, wax as a secondary gelator) may suit open-rack modules.
- **The tilted settling assay** of §3.5.2.
- **A longer field campaign:** several modules per treatment, a bare calibration day, sensors at the slab centre and on its outer face (H7c), contained slabs, an on-site pyranometer, full-size modules, and the C40 – wax margin as the primary outcome.

9. Acknowledgements

[FILL] We thank [supervisor name] for supervision and [chemistry teacher name] for supervising all heated laboratory work (§3.10); the Chemistry Laboratory of Cadet College Hasanabdal for bench and oven time; [GIKI department / facility and staff names] for DSC, FTIR and SEM access; and [anyone else who gave lab time, materials, photographs or review]. Materials were obtained from Bioshop PK (lauric acid) and Aogubio Co., Ltd. (rice bran wax, batch PR 1798).

Appendix A. Panel model equations

The module (glass, cells, backsheet) is one node at T_{m} with areal heat capacity C''; the composite is a second node with specific enthalpy H. For module area A,

$$C'' A \frac{\mathrm{d}T_m}{\mathrm{d}t} = (\tau\alpha)\, S(t)\, A - q_f - q_b,$$

$$q_f = h_w A\,(T_m - T_a) + \varepsilon_f \sigma A \left[F_s \left(T_m^4 - T_{\mathrm{sky}}^4\right) + F_g \left(T_m^4 - T_a^4\right)\right], \qquad h_w = 5.8 + 3.7\,V,$$

with $F_s = (1 + \cos\beta)/2$, $F_g = (1 - \cos\beta)/2$ for tilt β, $T_{\mathrm{sky}} = 0.0552\, T_a^{1.5}$ (Swinbank, 1963), and radiative temperatures in kelvin. The back-path heat flow is

$$q_b^{\mathrm{bare}} = B(T_m, \varepsilon_b), \qquad q_b^{\mathrm{PCM}} = \frac{kA}{L}(T_m - T_c), \qquad m \frac{\mathrm{d}H}{\mathrm{d}t} = q_b^{\mathrm{PCM}} - B(T_c, \varepsilon_c),$$

with composite mass $m = \rho AL$ and back boundary

$$B_{\mathrm{open}}(T, \varepsilon) = h_b A\,(T - T_a) + \varepsilon\sigma A\left(T^4 - T_a^4\right), \qquad B_{\mathrm{insulated}}(T) = U A\,(T - T_a), \quad U = \left[\frac{L_i}{k_i} + \frac{1}{h_b + 4\varepsilon_i \sigma T_0^3}\right]^{-1},$$

where $T_0 = 300$ K and $\varepsilon_i = 0.90$, giving $U \approx 1.5$ W/m²·K for 20 mm of expanded polystyrene. Temperature and liquid fraction follow from H (zero at the solidus) with the latent heat released linearly across $[T_{\mathrm{s}}, T_{\mathrm{l}}]$:

$$T_c = \begin{cases} T_s + H/c_p, & H \le 0, \\ T_s + \dfrac{H}{c_p + \Delta H/(T_l - T_s)}, & 0 < H < H_l, \\ T_l + (H - H_l)/c_p, & H \ge H_l, \end{cases} \qquad H_l = c_p(T_l - T_s) + \Delta H, \qquad f = \frac{T_c - T_s}{T_l - T_s} \text{ (clipped to [0, 1])}.$$

The synthetic climate is $T_a(t) = 22 + 7\sin[2\pi(t_h - 9)/24]$ °C and $S(t) = 800\ \sin[\pi(t_h - 6.5)/11.5]$ W/m² between 06:30 and 18:00, zero otherwise; efficiency is $\eta = \eta_{\text{ref}}\,[1 - \beta\,(T_m - 25)]$. Parameters are in Table 3.1.

Appendix B. Leak-test handling and decision rules

Paper conditioning (protocol). All papers dried together at 105 °C for 1 h, cooled in a desiccator and stored over silica gel; two blanks per run (three if conditioning is omitted), which must agree to ~0.01 g. *As run: papers used from the box, two blanks per run.*

Weighing. Balance warmed 30 min, levelled, zeroed, re-zeroed every ten weighings with drift logged; gloves and forceps throughout. Each mass is the mean of three separate placements (uncertainty ≈ ±0.006 g); a spread above 0.02 g stops weighing. After the hold, specimens cool 30 min in a desiccator and are weighed only once a co-located dummy with a sensor reads ambient, within 60 s of opening. *As run: 30 min on the bench, temperature not measured.*

Oven. Preheated ≥ 45 min; all ten positions loaded in under 30 s in the pre-generated random order; the loaded shelf photographed as the position map; the door not reopened for two hours.

Precision. LOD one balance increment (mean 0.21 % of m_0), LOQ three (mean 0.64 %); leakage reported to 0.1 %. An earlier fixed 0.4 % "below detection" threshold, derived for 5–7 g specimens, was replaced by these per-specimen limits when the 50 °C block showed the specimens were lighter than nominal. The |Leakage − Transfer| > 0.4 % flag is about two increments wide on the specimens cast; on disagreement the checks are, in order: large closure (evaporation; consult the sealed control), liquid off the paper, blank correction applied, anything weighed warm.

Rules fixed and dated before the first run.

Rule	Decision
Collapse	Specimen not separable from its paper intact ⟶ record "collapsed", leave Leakage blank, record a definitive **fail** at that temperature. Never reported as a number.
Oven excursion	Displayed temperature outside setpoint ± 1 °C at any checkpoint, or any alarm / unplanned door opening ⟶ **void the whole run** (all seven specimens). No salvage of individual specimens.
Handling loss	Specimen dropped, chipped, or in contact with anything but its own paper ⟶ void that specimen, record why.
Defective specimen	Visible void, crack or skin defect ⟶ excluded before any run.
Blank drift	|Δm_blank| > 0.03 g ⟶ conditioning failed; void that run's cross-checks, keep primary leakage numbers, flag them as uncorroborated.
Replicates	All three reported. No outlier dropped without a documented physical cause from the rules above.

Stopping point. The 50 °C series was completed and examined before committing material to 60 and 70 °C.

Appendix C. Samples requested for external analysis

DSC. Core set: raw RBW (batch PR 1798), pure lauric acid and composites at 30, 50 and 70 wt %; extended to 20, 40, 60 and 80 wt % and cycled C40 if instrument time allows. Pans loaded molten, so that a 5–10 mg sample represents a composite that is heterogeneous when solid. Programme 20 ⟶ 90 ⟶ 20 ⟶ 90 °C at 1–2 °C/min under N_2 (slow, because the features lie a few degrees apart). Recorded per sample: mass, pan type, atmosphere, onset, peak and end temperatures and enthalpy of every transition; supercooling = melting onset − crystallisation onset, compared at matched mass and cooling rate.

FTIR (ATR). Pure lauric acid, raw RBW, composites at 30, 50 and 70 wt % at minimum (all seven if time allows) and cycled C40 if submitted, as a composition series. Bands: acid C=O near 1700 cm^{-1}, wax-ester C=O near 1735 cm^{-1}, O–H 2500–3300 cm^{-1}. Raw spectra exported.

SEM. Sputter-coated fresh fracture surfaces of raw RBW, one 20–30 wt % and one 70–80 wt % composite, and uncycled and cycled C40 if submitted. One spare C40 specimen is kept unheated as the reference, and the cycled piece is taken only after its final weighing. Network density and needle length measured in ImageJ ($n \geq 30$ features).

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